Power and Its Shadow: When Unanimity Serves the Hegemon

Manoel Galdino (University of São Paulo; mgaldino@usp.br)

2026-08-22

Abstract

International organizations combine formally equal votes with large inequalities in bargaining power. This paper isolates one channel through which that combination can favor a hegemon. In a two-round benchmark, only weak states propose and a hegemon privately knows whether its disagreement payoff is low or high. Majority permits a proposer to replace the hegemon with uninformed weak-state votes, while unanimity makes its privately informed approval indispensable. Comparing each private game with its public-type counterpart separates the value of a necessary vote from type-specific informational rent. The institutional ranking is conditional and the maintained pure-strategy correspondence is empty in some belief regions. We then add an earlier, mandatory stage in which the informed hegemon proposes. Public agenda power can favor either rule: majority has an advantage when the outside option is low, while unanimity can gain when the outside option makes majoritarian delay attractive. Under private information, the structural causal contrast of agenda is a correspondence and decomposes as $T = D + I$, where D is the complete-information agenda effect and I is its interaction with informational rent. Agenda weakly benefits both types under unanimity wherever both arms exist; no general sign is imposed under majority. The analysis preserves linked type vectors, empty cells, and equilibrium multiplicity. It shows how veto indispensability, agenda power, and private information jointly shape the distributive value of formally equal approval rules.

1 Introduction

At the height of American power, the United States led the creation of the World Trade Organization (WTO). One would expect that bargaining outcomes in the trading system would reflect the asymmetry of power in the international system. Surely, International Organizations created by the hegemon would reflect such asymmetries in their rules. And yet, the WTO was created on the basis of equality rather than power: decisions by consensus meant that each member had equal veto power, and the US had no agenda power. Why would a hegemon help create an organization in which every state would formally be its equal?

The standard explanation is that equality under the law does not mean equality in practice. Coercion and Information institutions rule the negotiation process and shape the results, favoring the powerful over the weak (Steinberg 2002). The US and the (then) European Community drafted the negotiation texts used by all, offered market access and exit options to alter other states' alternatives, assembled linked packages, and gathered information through successive proposals and objections.

The best example is the process that ended with the creation of the WTO. At the close of the Uruguay Round, they threatened states that did not accept the creation of the WTO with withdrawal from the GATT 1947. The status quo was no longer available for such states. This episode shows how material power can enter a formally egalitarian institution through the agenda, the contract space, and the value of disagreement.

From the perspective of the distributive rationalist tradition, which focuses on bargaining theories of multilateral bodies (Voeten 2019), it is hard to reconcile this explanation with a rationalist framework. Consider the standard Baron and Ferejohn framework (Baron and Ferejohn 1989). Under open rule and equal recognition of the probability of proposal, combined with unanimity, there is no asymmetry of power and, as a result, no inequality of outcomes (Kalandrakis 2006). Concretely, if the US or the so-called Quad (US, EC, Japan, and Canada) had no formal agenda power, any country could propose a deal at any stage of a bargaining round. In fact, many coalitions did that during the Uruguay Round; for example, the agro-export group of countries called the Cairns Group (Cairns Group 1987). So, either a hegemon like the US made a blunder, or there is a need to further explain the institutional architecture of institutions like the OMC from a rationalist point of view, much like Fearon (1995) did for war.

More recently, the formal literature has investigated multilateral settings in which a single uninformed player is the sole proposer and multiple symmetric responders privately know their own types. On the one hand, they highlighted how the uninformed player may optimally choose to transfer informational rents as insurance against rejection (Glynia, Thum, and Xefteris 2026). On the other hand, they show that informed responders can likewise obtain informational rents by acting tough, since rejection under unanimity can improve their continuation offers (Piazolo and Vanberg 2025).

However, their models do not study settings in which power, rather than weakness, is concentrated in a single player, a key theme in International Relations. There is also no explanation of what happens to informational rent when the same actor holds both private information and proposal power, nor of the role of screening by uninformed weak players versus signaling by the informed player when proposing. Most importantly, in these models, changing the voting rule changes how many informed voters the proposer must buy and therefore the relative appeal of pooling and screening, but every winning coalition still includes an informed responder. They leave open a more fundamental question: what happens to the informed actor's bargaining power and informational rents when a minimum winning coalition can be formed entirely from uninformed voters? This question is central to understanding the comparison between majority and unanimity from the perspective of the most powerful actor.

The current paper fills this gap with a general n-player bargaining game in which power is concentrated in a single player, the hegemon, while all other players are symmetric and weak. As in the existing literature, the privately informed actor can be either a high or a low type. I first investigate what happens when even the powerful actor is stripped of proposal power. I then reintroduce proposal power by allowing the hegemon to make the first proposal, as often occurs in international organizations.

I show that..

Section 3 works a numerical case. Section 4 defines the game and solution concept. Section 5 solves the complete-information benchmarks, then the private games, and finally the informational rents and their difference across voting rules. Section 7 interprets the essential-input mechanism and its limits. The appendices give the proofs, the endpoint conventions, the exact correspondence tables, and the worked values.

2 Literature and contribution

2.1 Formal equality and informal power in international institutions

The paper begins from an established problem in international political economy. International organizations often give states formally equal rights while operating in a system with large disparities in markets, outside options, and coercive capacity (Krasner 1983; Koremenos, Lipson, and Snidal 2001). Research on informal governance shows how powerful states can preserve formally neutral rules while influencing agendas, staffing, enforcement, and the alternatives available to weaker members (Steinberg 2002; Stone 2011). The result is neither formal equality translated mechanically into equal influence nor material power operating wholly outside institutions. Power works through the bargaining environment in which formal rules are used.

Steinberg's study of the GATT and WTO supplies the closest substantive account (Steinberg 2002). He distinguishes law-based bargaining, in which members take sovereign-equality procedures seriously, from power-based bargaining that invisibly weights outcomes toward the United States and the European Community. His historical evidence covers several practices: Green Room and Quad agenda setting, control of drafts and packages, market-access leverage, sanctions and forum exit, side payments and issue linkage, successive probes of state preferences, and legitimacy generated by an appearance of consent. These practices do not all constitute separate mechanisms. They can be organized by the strategic object each one changes.

Table 1: Steinberg's practices and their strategic counterparts

Observed practice	Strategic object	Relation to this paper
Green Room, Quad, drafting, difficult-to-amend packages	Recognition, proposal, and amendment protocol	Familiar agenda power; deliberately removed from the baseline hegemon
Market size, sanctions, alternative forums, withdrawal from GATT 1947	Disagreement and continuation payoffs	Microfoundation for asymmetric and possibly privately known $o \in \{\ell, h\}$
Side payments, exceptions, transition periods, issue linkage, single undertaking	Feasible contract space and closure rule	Motivation for a divisible package; single undertaking is distinct from destruction of the old fallback
Proposals, objections, working groups, and plenary probes	Information structure	Steinberg emphasizes information flowing from weak members to agenda setters; the model studies private information held by the hegemon
Legitimacy, implementation, and reputational concerns	Future participation and enforcement	Complementary mechanisms outside the paper's two-round baseline

This reconstruction clarifies the paper's counterfactual. The model accepts the asymmetry of disagreement values and the availability of compensating concessions, both of which Steinberg helps motivate. It then removes the hegemon's exclusive proposal power. That restriction does not assert that agenda power was absent from the WTO. It permits a cleaner question: can a formally equal approval rule give a privately informed hegemon a distributive advantage even when weak states control proposals?

2.2 Proposal rights and multilateral bargaining

Baron and Ferejohn provide the canonical framework for that question (Baron and Ferejohn 1989). Their model places recognition, proposals, voting, and continuation values inside a sequential non-cooperative game. This move makes legislative outcomes depend on the rules governing both agenda formation and approval, rather than on voting power alone. It also establishes the vocabulary used here: recognized proposers divide a fixed surplus, build winning coalitions, and compare immediate agreement with continuation.

Kalandrakis isolates the distributive force of recognition more sharply (Kalandrakis 2006). For monotonic voting rules and arbitrary discount factors, appropriate proposal probabilities can support any division of the surplus as stationary equilibrium payoffs. Proposal rights can therefore dominate the political power attributed to votes or patience. This is the benchmark the present paper switches off. The hegemon is never recognized in the baseline, so a favorable payoff cannot be attributed to its probability of proposing. Work on proposal and veto rights and the broader legislative- bargaining literature further shows that approval thresholds, continuation values, and recognition jointly determine who is bought into a coalition (Winter 1996; McCarty 2000; Eraslan and Evdokimov 2019).

Heterogeneity in disagreement values supplies the complete-information bridge. Miller, Montero, and Vanberg show that a player with an expensive known vote can benefit under unanimity and be excluded from a minimal winning coalition under majority (Miller, Montero, and Vanberg 2018). That is a public-type effect of pivotality and coalition composition. The contribution here begins after this benchmark by asking what private information adds to the same institutional comparison.

2.3 Private information and the price of approval

Piazolo and Vanberg offer the closest dynamic comparison (Piazolo and Vanberg 2025). A fixed, uninformed proposer bargains for two periods with responders who privately know their disagreement values. Rejection in the first period can signal that a responder is expensive. Under unanimity, that reputation improves the responder's continuation offer; under majority, it can lead to exclusion from the next coalition. Their result establishes the comparative root on which this paper builds: toughness is rewarded when a vote is indispensable and punished when the voter can be replaced. It also predicts visible bargaining friction, because rejection carries the signal.

Glynia, Thum, and Xefteris close the dynamic channel (Glynia, Thum, and Xefteris 2026). Their proposer makes one offer without knowing the acceptance costs of symmetric responders. The relevant choice is between a costly package that insures against high types and a cheaper gamble that buys only enough low-cost votes. Unanimity can induce the safe package, while majority can induce the gamble. This mechanism can raise approval under unanimity or raise total transfers under majority, reversing familiar rankings without learning through delay.

These papers establish mechanisms that the present analysis should not claim as new. Piazolo and Vanberg show how the voting rule changes the signaling value of rejection. Glynia, Thum, and Xefteris show how it changes the proposer's insurance problem. The present paper changes the organization of private information. One asymmetric actor, the hegemon, is informed; the weak states that can substitute for it are not. Majority can therefore bypass the only private information relevant to the proposed payment. Unanimity makes that information unavoidable.

The distinction produces two results absent from the neighboring models. First, the approval rule acts as an on-off switch for the informed participant rather than only changing the price of one informed vote relative to another. Second, the public-type benchmark permits a type-specific decomposition between the value of a necessary vote and the additional rent created by uncertainty. In the pooling region, that rent can arise through immediate agreement, without an on-path rejection. The paper therefore studies who receives the surplus when the only privately informed actor moves from replaceable to indispensable.

Related models of incomplete-information bargaining examine majority-only screening, cheap talk, reputational toughness, and informational loss (Tsai 2009; Tsai and Yang 2010; Chen and Eraslan 2013, 2014; Ma 2023). Informational voting models ask how votes aggregate signals distributed across voters (Feddersen and Pesendorfer 1998). Here weak states hold no private signals about the hegemon. The institutional comparison operates through coalition composition and the price of one actor's privately known participation constraint.

3 A working numerical illustration

Consider one hegemon and four weak states. The unit pie is bargained over with discount factor 0.9. The two disagreement payoffs are 0.10 and 0.35. In the terminal unanimity round, a proposer chooses between an offer accepted only by the low outside option and an offer accepted under either outside option. The cutoff probability of the high outside option is

$$p^* = \frac{0.35 - 0.10}{1 - 0.10} = 0.2778.$$

At a belief of 0.20, the proposer offers 0.10 and accepts the risk of failure when the outside option is high. At a belief of 0.35, it offers 0.35 and both outside-option values accept. This is a working numerical illustration, not an empirical calibration. The full two-round game adds the option to buy weak-state votes and the continuation created by a failed first ballot.

4 The model

4.1 Players, information, and proposals

There is one hegemon, H, and $m \geq 3$ weak states, $W = \{1, \ldots, m\}$, for a total of $m + 1$ states. Nature selects H's terminal disagreement payoff $o \in \{\ell, h\}$, observed only by H, with $\Pr(o = h) = p \in [0, 1]$. The two possible values satisfy

$$0 < \ell < h < 1.$$

The game has two rounds. In each round, one weak state is recognized uniformly to propose; H is never a proposer. The two recognition draws are independent and made with replacement. Every weak state remains eligible in Round 2, including the Round-1 proposer. A proposal by weak state i is the allocation vector

$$x = (x_H, (x_j)_{j \in W}) \in \mathcal{X}, \qquad \mathcal{X} = \left\{ x : 0 \leq x_H \leq \bar{x}_H,\ x_j \geq 0,\ x_H + \sum_{j \in W} x_j \leq 1 \right\},$$

where $h \leq \bar{x}_H \leq 1$. Payments cannot be negative and side payments outside the proposed package are unavailable. Here x_H is the institutional concession assigned to H, and x_j is weak state j's share, including the proposer when $j = i$. The fixed unit pie is exhausted on the equilibrium path. Agreement provides H no intrinsic benefit beyond x_H, which isolates informational concessions and pivotality from any direct value of agreement.

4.2 Ballots, timing, and payoffs

The proposer is counted as voting yes. All other states vote simultaneously; the complete vote vector and outcome become public only after all votes have been cast. Define

$$k = \left\lfloor \frac{m+1}{2} \right\rfloor.$$

Under majority, passage requires k additional yes votes beyond the proposer's; unanimity requires every state's yes vote. The ballot protocol is otherwise identical across rules.

If a proposal passes, every stated allocation is implemented. If H votes yes, it receives x_H. If H votes no but a majority passes the proposal without it, its general payoff is $x_H + o$. This formula is part of the complete game because the proposed institutional term x_H is implemented even without H's approval. In every equilibrium exclusion derived below, however, the proposer chooses $x_H = 0$; the hegemon therefore receives exactly o, never a positive concession on top of it.

If nothing passes in Round 1, no payoff is realized then and the game proceeds to Round 2. Round-2 payoffs are multiplied by $\beta \in (0, 1)$ when evaluated in Round-1 units. If nothing passes in Round 2, each weak state receives zero and H receives o in Round-2 units. A no vote is only a ballot action; it does not remove a player or trigger disagreement by itself. [AUTHOR: P1] The delayed terminal disagreement payoff represents the cost of prolonging international negotiations.

Table 2: Transitions and payoffs in the essential-input game

Event	Weak-state payoffs	H votes yes	H votes no
Proposal passes	Proposed x_j, including x_i	x_H	$x_H + o$
Round-1 proposal fails	No current payoff; proceed to Round 2	$\beta C_H(\mathfrak{h}^Y)$	$\beta C_H(\mathfrak{h}^N)$
Round-2 proposal fails	Zero	o	o

Here $\mathfrak{h}^Y$ and $\mathfrak{h}^N$ denote the distinct public histories in which H voted yes or no, and $C_H(\mathfrak{h}^a)$ denotes H's Round-2 continuation value after history $\mathfrak{h}^a$. These continuation values need not be equal because the public vote can change beliefs before Round 2.

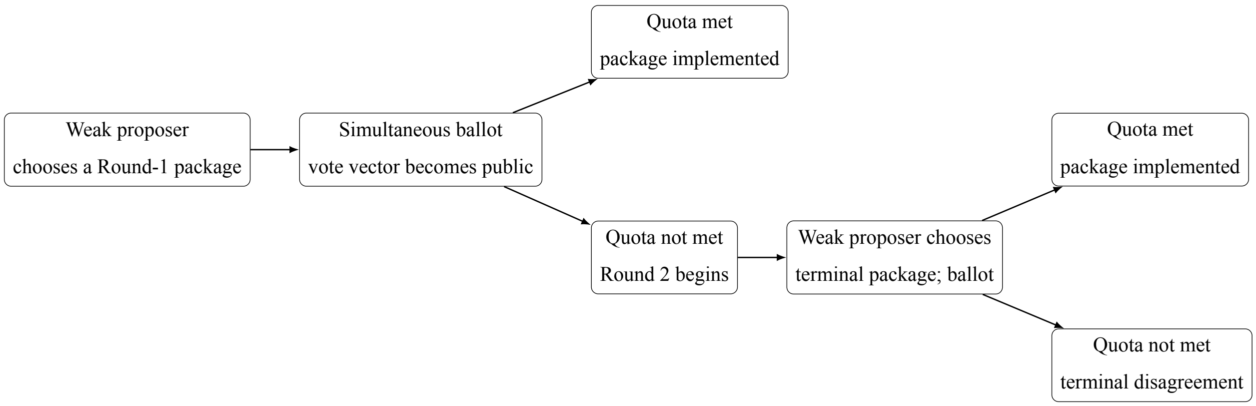

Figure 1: Sequence of proposals and ballots. Within each ballot, responding states vote simultaneously. A failed first-round proposal continues to the terminal round; a no vote alone does not end the game.

4.3 Solution concept

We use perfect Bayesian equilibrium with pure ballot strategies and three declared disciplines. First, an action by a player who does not know o does not change beliefs about o. After an action by H, beliefs follow Bayes' rule whenever possible and otherwise satisfy structural consistency. Second, a weak responder evaluates yes and no as if its own vote were pivotal. Third, a voter accepts when exactly indifferent in expected value. Among proposal choices that give the proposer the same expected payoff, the selected proposal minimizes H's expected payoff.

Endpoint beliefs preserve the support of the prior. If $p = 0$, the posterior assigns zero probability to h at every history; if $p = 1$, it assigns probability one to h. For $0 < p < 1$, an action by H that has zero probability under both outside-option values may support any posterior in $[0, 1]$. This is a condition of our solution concept. It follows the no-signaling logic used in multistage games (Fudenberg and Tirole 1991); support preservation is analogous to, but narrower than, the never-dissuaded discipline discussed by Osborne and Rubinstein (1990). Tremble consistency motivates the restriction but is not invoked as a theorem for degenerate priors (Kreps and Wilson 1982).

Table 3: Scope of the formal results

Element	Maintained scope
Players and horizon	One hegemon, $m \geq 3$ weak states, two rounds
Agenda	Only weak states propose in both rounds
Voting	Simultaneous public ballots; majority or unanimity
Information	Private outside option $o \in \{\ell, h\}$; support-preserving endpoints
Payoffs	Unit pie, $0 < \ell < h < 1$, $0 < \beta < 1$; no intrinsic agreement benefit for H
Equilibrium	Perfect Bayesian equilibrium in pure ballot strategies with the declared voting and proposal tie-break rules

4.4 Separate agenda-stage extension

The benchmark above remains the paper's primary model. After solving it, we consider a separate extension in which the informed hegemon must make an earlier proposal at date A. Passage implements that proposal immediately; failure enters Round 1 of the benchmark and transports its complete continuation by exactly one factor β. The extension gives H no option to skip the proposal while retaining a date-A continuation. Section 6 states the economic results, and Appendices E–F preserve the complete equilibrium correspondences and date conventions.

5 Results

The solution proceeds by backward induction. We begin with public types because that benchmark separates the value of a necessary vote from the value of private information. We then solve the private terminal games, the private first-round games, and finally subtract the public benchmark component by component.

5.1 Complete-information benchmark

With a public type, no belief or signaling issue remains. The proposer compares the price of H's vote with the price of enough weak-state votes. In the terminal round, the relevant price of H is o. In the first round, it is βo, because rejection leads to the terminal round. Under majority, a weak-state vote costs β/m in first-round units. Under unanimity with public disagreement payoff o, each weak responder's continuation price is $\beta(1-o)/m$.

Proposition 5.1 (Public-type benchmark)**.** *Fix a public type with disagreement payoff o. In the terminal round, majority passes a proposal without H, assigns $x_H = 0$, and gives the weak proposer the whole unit pie; H receives o. Terminal unanimity passes with $x_H = o$, and the weak proposer retains $1-o$.*

In Round 1, unanimity passes immediately with $x_H = \beta o$. Majority includes H if $o \leq 1/m$ and excludes it if $o > 1/m$. At equality the proposal including H is selected. Consequently, the public Round-1 payoff of H is

$$v_M^B(o) = \begin{cases} \beta o, & o \leq 1/m, \\ o, & o > 1/m, \end{cases} \qquad v_U^B(o) = \beta o.$$

Table 4: Complete-information outcomes by round and voting rule

Round	Rule	Proposed payments	Weak proposer	H
2	Majority	$x_H = 0$; weak responders get zero	1	o
2	Unanimity	$x_H = o$; weak responders get zero	$1 - o$	o
1	Majority, $o \leq 1/m$	$x_H = \beta o$; $k-1$ responders get β/m	$1 - \beta o - (k-1)\beta/m$	βo
1	Majority, $o > 1/m$	$x_H = 0$; k responders get β/m	$1 - k\beta/m$	o
1	Unanimity	$x_H = \beta o$; every responder gets $\beta(1-o)/m$	$1 - \beta o - (m-1)\beta(1-o)/m$	βo

At $o = 1/m$, buying H's vote and buying the substitute weak-state vote cost the proposer the same amount. The proposal tie-break selects inclusion because it gives H the lower payoff $\beta o < o$.

5.2 Private information in the terminal round

Let

$$p^* = \frac{h - \ell}{1 - \ell}.$$

Under terminal majority, H's vote is unnecessary and the belief never enters the proposer's choice. Under terminal unanimity, the proposer compares a low offer that succeeds only when the type is low with a high offer that both types accept.

Proposition 5.2 (Private terminal games)**.** *Under majority, the unique equilibrium outcome assigns $x_H = 0$, pays no responding weak state, and gives the full pie to the proposer. The proposal passes without H; a hegemon with outside option o receives o, and a weak state receives $1/m$ before recognition.*

Under unanimity, if $p \leq p^*$, the proposer offers $x_H = \ell$. The low outside-option type accepts and the high outside-option type rejects, so the proposal succeeds with probability $1-p$. If $p > p^*$, the *proposer offers $x_H = h$ and both types accept. At $p = p^*$, the low offer is selected.*

The cutoff equates $(1-p)(1-\ell)$, the proposer's payoff from the low offer, with $1-h$, its payoff from pooling. Notice that no discount factor appears inside this terminal comparison.

5.3 Private majority in Round 1

Let $w = \beta/m$. The proposer's payoffs from the three undominated outcome classes are:

$$\begin{aligned} \Pi_E &= 1 - kw, \\ \Pi_S(p) &= (1-p)\left[1-(k-1)w-\beta\ell\right] + pw, \\ \Pi_P &= 1-(k-1)w-\beta h. \end{aligned}$$

Exclusion buys k weak responders at price w each and sets $x_H = 0$. Screening buys $k-1$ weak responders and offers $\beta\ell$: the low outside-option type accepts and the high outside-option type delays. Pooling buys $k-1$ weak responders and offers βh, which both types accept. Deliberate delay is strictly worse than exclusion because $1-\beta(k+1)/m > 0$.

When $\ell < 1/m$, define

$$p_{S=E} = \frac{\beta(1/m-\ell)}{\beta(1/m-\ell)+1-\beta(k+1)/m}.$$

When $h < 1/m$, define

$$p_{S=P} = \frac{\beta(h-\ell)}{1-\beta\ell-\beta k/m}.$$

Proposition 5.3 (Private majority correspondence)**.** *For $m \geq 3$, the equilibrium outcome class is:*

1. *If $h < 1/m$, screening for $p \leq p_{S=P}$ and pooling above it.*
2. *If $\ell < 1/m < h$, screening for $p \leq p_{S=E}$ and exclusion above it.*
3. *If $1/m < \ell < h$, exclusion for every p.*
4. *If $\ell = 1/m < h$, screening at $p = 0$ and exclusion for $p > 0$.*
5. *If $\ell < h = 1/m$, screening for $p \leq p_{S=E}$. Above that cutoff, exclusion is selected when $(1-p)\ell + ph < \beta h$, pooling is selected when the inequality is reversed, and the entire*

proposal segment joining the two survives when the two expressions are equal.

At every cutoff involving screening, screening is selected. Permuting the identities of identically paid weak responders generates additional equilibria but does not change H's payoff vector or the outcome class. Payoffs of weak states identified by label can be permuted and, on the residual proposal family, vary with the common proposal weight.

The last segment is genuine multiplicity. Its mixing weight is a probability over pure proposals in the proposer's strategy, and the same weight must be used jointly for payoffs and outcomes. It is not a license to combine componentwise minima and maxima into an unattained rectangle.

5.4 Private unanimity in Round 1

Let

$$w_\ell^U = \frac{\beta(1-\ell)}{m}, \qquad w_h^U = \frac{\beta(1-h)}{m}.$$

At $p = 0$, support preservation prevents an impossible high type from being reintroduced by an off-path belief. The proposer can therefore offer the low threshold $\beta\ell$, pay each responding weak state its continuation w_ℓ^U, and keep $w_\ell^U + 1 - \beta$. When $p > p^*$, the continuation itself pools, so the proposer offers βh, pays each responder w_h^U, and keeps $w_h^U + 1 - \beta$.

The intermediate cell fails for a different reason. A feasible deviation can pay every weak responder enough to force a yes vote. Once all weak votes are yes, no pure contingent voting profile for the two possible types of H survives: pooling yes is rejected by the high type, pooling no is overturned by the low type at equality, and either separating profile gives one type a profitable imitation.

Proposition 5.4 (Private unanimity correspondence)**.** *Under unanimity:*

$$\begin{aligned}
p = 0 \quad &: \quad \textit{immediate low-type agreement with } x_H = \beta\ell,\ x_j = w_\ell^U,\ x_i = w_\ell^U + 1 - \beta; \\
0 < p \le p^* \quad &: \quad \textit{no perfect Bayesian equilibrium in pure ballot strategies}; \\
p^* < p \le 1 \quad &: \quad \textit{immediate pooling agreement with } x_H = \beta h,\ x_j = w_h^U,\ x_i = w_h^U + 1 - \beta.
\end{aligned}$$

The corresponding payoff vectors for H, ordered as low and high type, are $(\beta\ell, \beta h)$ at $p = 0$ and $(\beta h, \beta h)$ above p^*.

Remark (Pure-strategy scope). The middle cell establishes nonexistence only within the paper's maintained space of pure ballot strategies. We do not derive, characterize, select, or compare equilibria involving mixed ballot strategies, and make no claim about existence under such strategies.

Table 5: Private-information Round-1 equilibrium correspondences

Rule	Parameter cell	Selected outcome class	H payoff vector
Majority	$h < 1/m$	Screening for $p \leq p_{S=P}$; pooling above	$(\beta\ell, \beta h)$ or $(\beta h, \beta h)$
Majority	$\ell < 1/m < h$	Screening for $p \leq p_{S=E}$; exclusion above	$(\beta\ell, \beta h)$ or (ℓ, h)
Majority	$1/m < \ell$	Exclusion for every p	(ℓ, h)
Majority	$\ell = 1/m < h$	Screening at $p = 0$; exclusion for $p > 0$	$(\beta\ell, \beta h)$ or (ℓ, h)
Majority	$\ell < h = 1/m$	Screening through $p_{S=E}$; above, exclusion, pooling, or their exact proposal segment under the tie-break	$(\beta\ell, \beta h)$, (ℓ, h), $(\beta h, \beta h)$, or the linked segment
Unanimity	$p = 0$	Immediate low-type agreement	$(\beta\ell, \beta h)$
Unanimity	$0 < p \leq p^*$	No PBE in pure ballot strategies	$\emptyset$
Unanimity	$p^* < p \leq 1$	Immediate pooling agreement	$(\beta h, \beta h)$

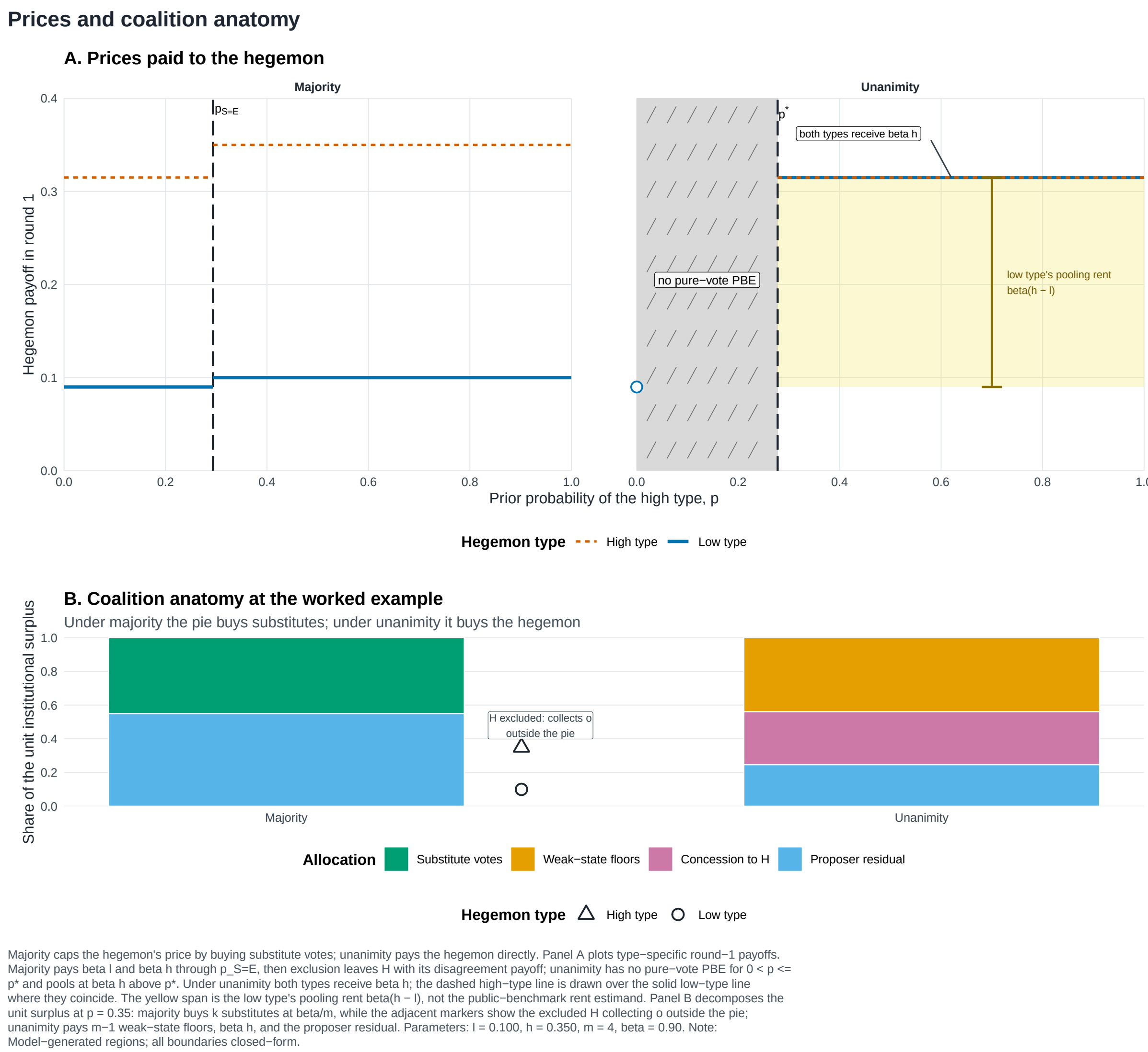

Figure 2: The price and coalition anatomy of the private games. Majority can replace the hegemon's vote with an additional weak-state vote; unanimity cannot. Labels report first-round prices and preserve the distinct low-type, pooling, and exclusion outcomes.

5.5 Comparing the private games

The comparison below uses the same parameter point under both institutions and only cells in which both games have a pure-strategy equilibrium. The timing wedges are $(1-\beta)\ell$ and $(1-\beta)h$: exclusion realizes the disagreement payoff now, whereas inclusion can price a continuation that is discounted in Round-1 units. They are distinct from the discounted type gap $\beta(h-\ell)$ and the cross-date quantity $\beta h - \ell$.

Proposition 5.5 (Private institutional payoff contrast)**.** *Write* $V_U^B - V_M^B$ *as a low-type/high-type vector.*

1. *At* $p = 0$*, the vector is* $(0, 0)$ *if majority screens and* $(-(1-\beta)\ell, -(1-\beta)h)$ *if majority excludes.*
2. *For* $0 < p \leq p^*$*, the contrast is empty because private unanimity has no pure-strategy equilibrium.*
3. *For* $p > p^*$*, the vector is* $(\beta(h-\ell), 0)$ *if majority screens,* $(0, 0)$ *if majority pools, and* $(\beta h - \ell, -(1-\beta)h)$ *if majority excludes.*
4. *On the residual exclusion–pooling segment at* $h = 1/m$*, the exact attainable set is* $\{\lambda(\beta h - \ell, -(1-\beta)h) : \lambda \in [0, 1]\}$*, restricted to the proposal weights that survive the tie-break. The same* λ *binds payoffs and outcomes.*

Thus unanimity does not have a single unconditional payoff effect. Its effect depends on which substitute coalition disciplines the majority proposer and on which type is evaluated. Figure 3 maps these exact classes without averaging types.

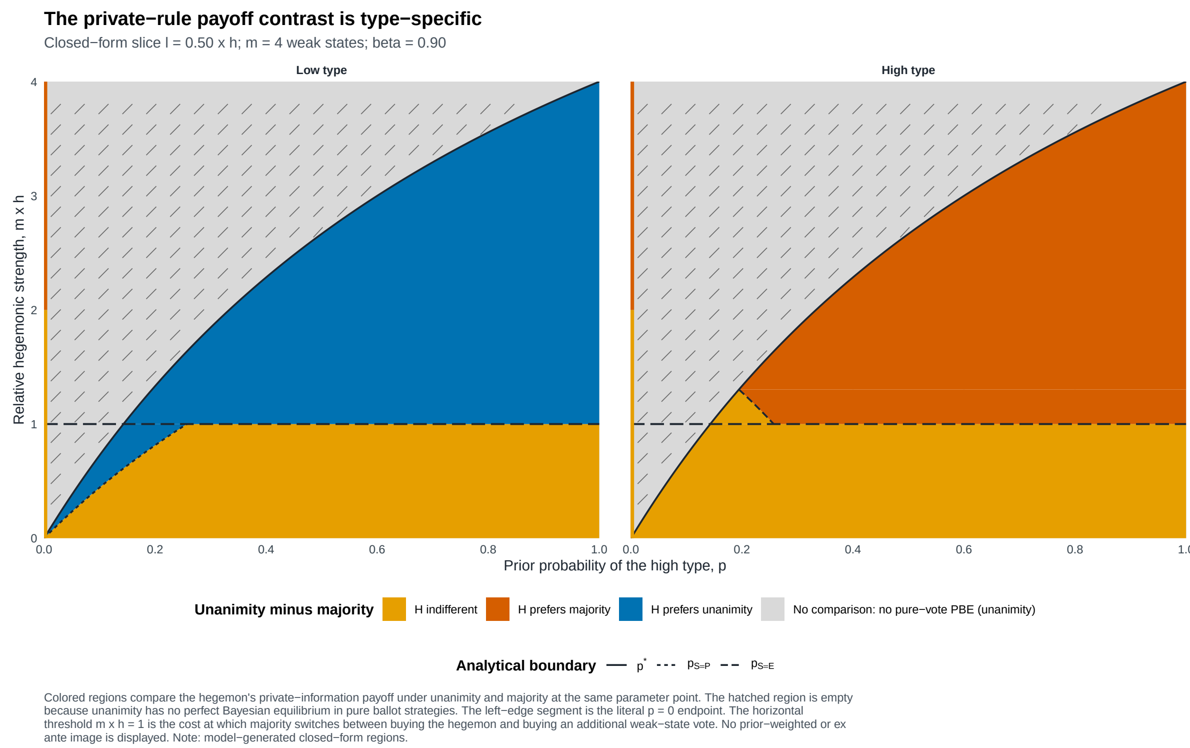

Figure 3: Private-game payoff contrast, unanimity minus majority, by hegemon type. Each cell is evaluated at the same parameter point. The hatched band marks parameter values for which private unanimity has no pure-strategy equilibrium and the contrast is therefore empty.

5.6 Informational rents and the difference of differences

For rule $g \in \{M, U\}$, define the type-contingent informational-rent correspondence

$$IR_g^B = V_g^B - v_g^B,$$

where the public benchmark is evaluated at the same type and the same remaining parameters. The institutional informational-rent contrast is

$$\Delta IR^B = IR_U^B - IR_M^B.$$

These are differences between correspondences. If a source correspondence is empty, so is the difference. Under multiplicity, subtraction preserves the entire attainable set and its common proposal weight.

We write the three public outside-option regions in words: both types are included when $h \le 1/m$; the low type is included and the high type excluded when $\ell \le 1/m < h$; and both types are excluded when $1/m < \ell$.

The rent comparison first asks what private information changes *within* each rule. Under unanimity, the public benchmark already buys the known type at its discounted threshold; private pooling therefore raises only the low type's payoff. Under majority, information matters only when the private equilibrium changes the public inclusion decision or the price paid to a type.

Proposition 5.6 (Informational rents by voting rule)**.** *Under unanimity,*

$$IR_U^B = \begin{cases} \{(0,0)\}, & p = 0, \\ \emptyset, & 0 < p \le p^*, \\ \{(\beta(h-\ell),0)\}, & p > p^*. \end{cases}$$

Under majority, the informational rent depends jointly on the public region and the private equilibrium class:

$$\begin{array}{rcl} \textit{both types included, screening} & : & (0,0), \\ \textit{both types included, pooling} & : & (\beta(h-\ell),0), \\ \textit{both types included, exclusion} & : & ((1-\beta)\ell,(1-\beta)h), \\ \textit{low included, high excluded, screening} & : & (0,-(1-\beta)h), \\ \textit{low included, high excluded, exclusion} & : & ((1-\beta)\ell,0), \\ \textit{both types excluded, exclusion} & : & (0,0). \end{array}$$

On the residual exclusion–pooling segment at $h = 1/m$, the exact set is $\{\lambda((1-\beta)\ell,(1-\beta)h) + (1-\lambda)(\beta(h-\ell),0) : \lambda \in [0,1]\}$, restricted to the segment that survives the proposal tie-break.

The difference of differences then removes majority's informational rent from unanimity's. Screening can leave a positive low-type contrast in the high-belief cell because unanimity pools. Exclusion instead compares the high threshold discounted to the present with the low current disagreement payoff, so its low-type sign is governed by $\beta h - \ell$.

Proposition 5.7 (Institutional informational-rent contrast)**.** *At $p = 0$, $\Delta IR^B = (0,0)$ when both* *types are included or both are excluded, while* $\Delta IR^B = (0, (1-\beta)h)$ *when the low type is included and the high type excluded; the high-type component there is off support but remains part of the type-contingent vector.*

For $0 < p \leq p^*$, $\Delta IR^B = \emptyset$.

For $p > p^*$, *the exact vectors are*

$$\begin{aligned}
\textit{both types included, screening} &: (\beta(h-\ell), 0), \\
\textit{both types included, pooling} &: (0, 0), \\
\textit{both types included, exclusion} &: (\beta h - \ell, -(1-\beta)h), \\
\textit{low included, high excluded, screening} &: (\beta(h-\ell), (1-\beta)h), \\
\textit{low included, high excluded, exclusion} &: (\beta h - \ell, 0), \\
\textit{both types excluded, exclusion} &: (\beta(h-\ell), 0).
\end{aligned}$$

On the residual segment, the exact attainable set is $\{\lambda(\beta h - \ell, -(1-\beta)h) : \lambda \in [0, 1]\}$ *over the surviving proposal weights.*

Table 6: Exact informational-rent correspondences and institutional contrast

Object	Cell	Exact set
IR^B_M	Both types included, screening	$\{(0, 0)\}$
IR^B_M	Both types included, pooling	$\{(\beta(h-\ell), 0)\}$
IR^B_M	Both types included, exclusion	$\{((1-\beta)\ell, (1-\beta)h)\}$
IR^B_M	Both types included, exclusion–pooling segment	$\{\lambda((1-\beta)\ell, (1-\beta)h) + (1-\lambda)(\beta(h-\ell), 0)\}$
IR^B_M	Low included, high excluded, screening	$\{(0, -(1-\beta)h)\}$
IR^B_M	Low included, high excluded, exclusion	$\{((1-\beta)\ell, 0)\}$
IR^B_M	Both types excluded, exclusion	$\{(0, 0)\}$
IR^B_U	$p = 0$	$\{(0, 0)\}$
IR^B_U	$0 < p \leq p^*$	$\emptyset$
IR^B_U	$p > p^*$	$\{(\beta(h-\ell), 0)\}$

Object	Cell	Exact set
ΔIR^B	$p = 0$, both types included	$\{(0,0)\}$
ΔIR^B	$p = 0$, low included, high excluded	$\{(0,(1-\beta)h)\}$
ΔIR^B	$p = 0$, both types excluded	$\{(0,0)\}$
ΔIR^B	$0 < p \le p^*$, every public region	$\emptyset$
ΔIR^B	$p > p^*$, both types included, screening	$\{(\beta(h-\ell),0)\}$
ΔIR^B	$p > p^*$, both types included, pooling	$\{(0,0)\}$
ΔIR^B	$p > p^*$, both types included, exclusion	$\{(\beta h-\ell,-(1-\beta)h)\}$
ΔIR^B	$p > p^*$, both types included, exclusion–pooling segment	$\{\lambda(\beta h-\ell,-(1-\beta)h) : \lambda \in [0,1]\}$
ΔIR^B	$p > p^*$, low included, high excluded, screening	$\{(\beta(h-\ell),(1-\beta)h)\}$
ΔIR^B	$p > p^*$, low included, high excluded, exclusion	$\{(\beta h-\ell,0)\}$
ΔIR^B	$p > p^*$, both types excluded, exclusion	$\{(\beta(h-\ell),0)\}$

In every segment row, $\lambda \in [0,1]$ is restricted to the proposal weights that survive the tie-break and binds all payoff and outcome components.

The sign pattern is transparent once each component is kept separate. Above the unanimity cutoff, the low type gains $\beta(h-\ell) > 0$ when majority screens because unanimity pools and pays the high threshold. The same screening comparison is zero at $p = 0$, while the middle cell is empty. When majority excludes, the low type's gain is $\beta h-\ell$, which is positive, zero, or negative according as βh is above, equal to, or below ℓ. The high type gains $(1-\beta)h > 0$ only when private majority screens even though public majority would exclude it; it loses $(1-\beta)h$ when public majority includes it but private majority excludes. When both private rules pool, the informational-rent contrast is exactly zero.

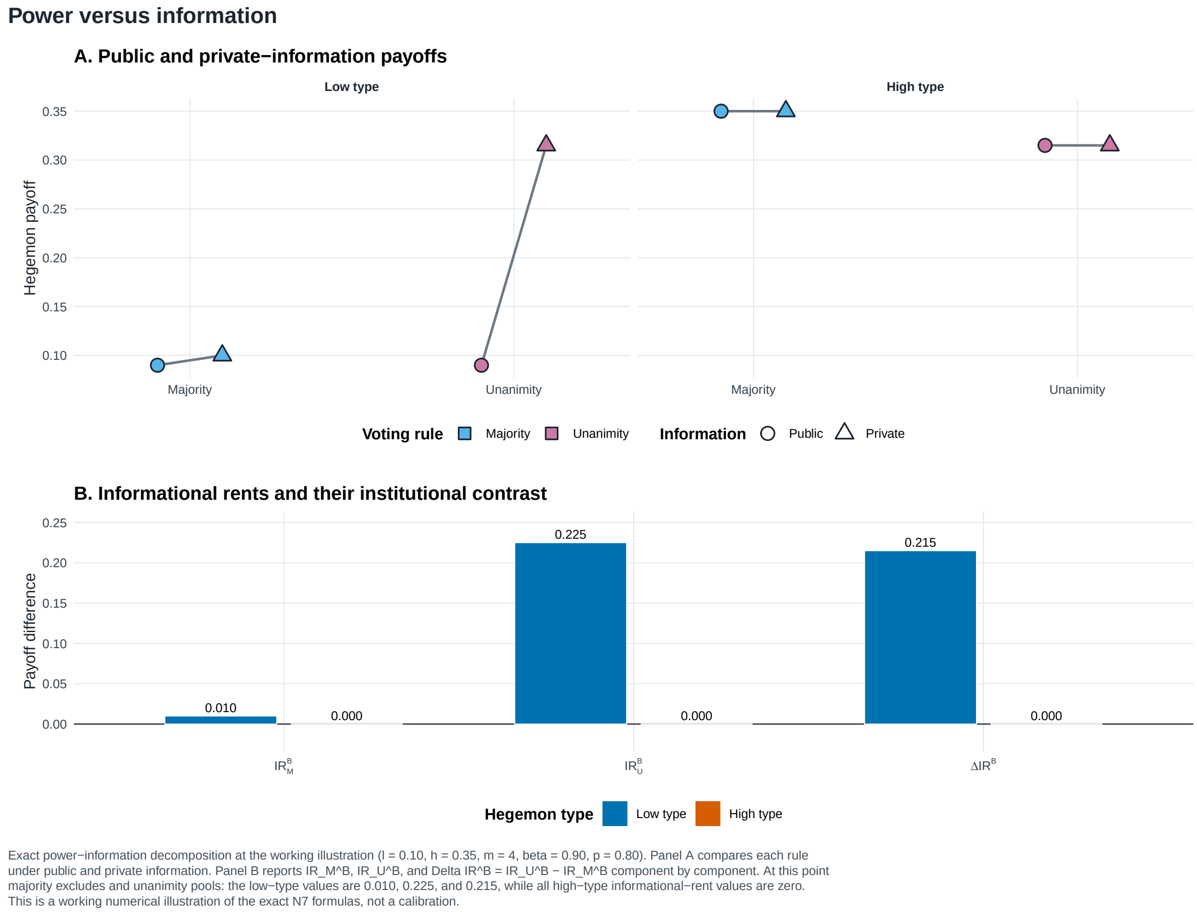

Figure 4: Power and information in the hegemon's payoff. The public benchmark captures the effect of making the hegemon's vote necessary when its type is known. Informational rent compares the private game with that benchmark, and ΔIR^B compares this rent across unanimity and majority without recombining types or selecting among multiple equilibria.

6 Agenda power and its informational shadow

The baseline removes an exclusive formal proposal privilege from H to identify informational power through pivotality. We now restore agenda power in a separate extension. Before the two-round game, at date A, the informed hegemon must propose a feasible division of the unit pie. If that proposal fails, the game enters Round 1 of the baseline. The continuation is therefore transported from Round 1 to date A by exactly one factor β. The proposal is mandatory: H cannot skip the agenda stage and obtain the continuation at date A.

Let

$$e = m - k, \qquad v_M^{\text{safe}} = 1 - \frac{k\beta}{m}.$$

Under majority, k is the number of weak votes that H must buy and e is the number of weak states that a minimal winning coalition can leave outside. The private games also retain the common off-path fiber (ρ, μ^{off}), where

$$\mu^{\text{off}} = \frac{p\rho}{1 - p + p\rho}$$

for an interior prior; at an endpoint, the fiber is $(*, p)$. Comparisons across rules always pair complete equilibrium records in the same economy and the same fiber. They do not splice type payoffs, proposals, or outcomes from different records.

6.1 Public agenda power

Suppose first that o is public before H proposes. Under unanimity, H buys every weak vote at its continuation price and agreement is immediate. Under majority, it buys a minimal coalition when passage is better than continuation, but a type with a sufficiently high outside option may deliberately induce rejection. The resulting payoffs at date A are

$$v_U^A(o) = 1 - \beta + \beta^2 o$$

and

$$v_M^A(o) = \begin{cases} 1 - \dfrac{k\beta(1 - \beta o)}{m}, & o \le 1/m, \\[2ex] \max\{v_M^{\text{safe}}, \beta o\}, & o > 1/m. \end{cases}$$

In the first majority branch, any lottery over minimal coalitions is retained. For $o > 1/m$, define

$$o_M^* = \frac{v_M^{\text{safe}}}{\beta} = \frac{1}{\beta} - \frac{k}{m} > \frac{1}{m}.$$

Agreement is immediate for $o < o_M^*$, any linked mixture of passage and delay is possible at equality, and every equilibrium delays for $o > o_M^*$. If $o_M^* \ge 1$, the last branch is outside the maintained domain $o \in (0,1)$. Unanimity does not inherit this delay: buying all votes beats rejection by exactly $1-\beta$.

Define the public institutional gap with the common orientation unanimity minus majority,

$$\Delta v^A(o) = v_U^A(o) - v_M^A(o).$$

Then

$$\Delta v^A(o) = \begin{cases} -\beta \dfrac{e}{m}(1-\beta o), & o \le 1/m, \\ \beta\left(\beta o - \dfrac{e}{m}\right), & o > 1/m \text{ and } \beta o \le v_M^{\text{safe}}, \\ (1-\beta)(1-\beta o), & o > 1/m \text{ and } \beta o \ge v_M^{\text{safe}}. \end{cases}$$

The last two expressions coincide at $\beta o = v_M^{\text{safe}}$. The first branch is negative; for $o > 1/m$, the sign of $\Delta v^A(o)$ is the sign of $\beta o - e/m$. Agenda power under public information thus need not favor unanimity. If $\beta h < e/m$, majority gives both types a strictly larger payoff; equality yields a weak ranking. Above that boundary, the public component can itself point toward unanimity.

Panel A: public institutional gap

$$\Delta v^A(o) = v_U^A(o) - v_M^A(o)$$

$$o \le 1/m \;:\; \Delta v^A(o) < 0,$$
$$o > 1/m \;:\; \operatorname{sgn} \Delta v^A(o) = \operatorname{sgn}(\beta o - e/m).$$

Public information favors majority below the boundary and can favor unanimity above it.

Panel B: an informational reversal

Low type	M	U	$U-M$
Public	.5905	.1810	-.4095
Private	.5905	.7290	+.1385
Information	.0000	.5480	+.5480

Figure 5: Public power and an informational reversal. Panel A reports the exact sign boundary for $\Delta v^A(o)$. Panel B uses one linked member with $m = 4$, $\beta = .90$, $\ell = .10$, $h = .90$, and $p = .95$; the unanimity value is the lower endpoint of its admissible pooling interval. The numbers are a theoretical illustration, not an empirical calibration or a global ranking of the correspondence.

6.2 Private agenda power and institutional comparison

Let $V_g^A = (V_g^A(\ell), V_g^A(h))$ denote the linked type-payoff vector generated by one complete private equilibrium record under rule $g \in \{M, U\}$. The majority correspondence contains pure pooling, separation, agreement, and deliberate-delay classes as well as Borel mixtures. It is not replaced by a selected equilibrium or by componentwise envelopes. Under unanimity, the exact payoff fibers can be written directly with the public value function $v_U^A(o) = 1 - \beta + \beta^2 o$:

$$V_U^A = \begin{cases} (v_U^A(\ell), \max\{v_U^A(\ell), \beta^2 h\}), & p = 0, \\ (v_U^A(\ell), v_U^A(\ell)), & 0 < p \le p^*,\ v_U^A(\ell) \ge \beta^2 h,\ \mu^{\text{off}} = 0, \\ \{(u,u) : u \in [\max\{v_U^A(\ell), \beta^2 h\}, v_U^A(h)]\}, & p^* < p < 1,\ \mu^{\text{off}} = 0, \\ (v_U^A(h), v_U^A(h)), & p^* < p < 1,\ \mu^{\text{off}} \in (p^*, 1], \\ (v_U^A(h), v_U^A(h)), & p = 1, \\ \emptyset, & \text{otherwise.} \end{cases}$$

Here and below $\emptyset$, also reported as `none` in the source correspondences, means an empty correspondence under the maintained pure-PBE M/S/B architecture. It is not a zero effect or a claim about mixed equilibria outside that architecture.

Institutional comparison begins with the exact fiber product

$$\mathscr{J}_A^{\text{bind}}(\mathbf{d}, \rho, \mu^{\text{off}}) = \mathscr{B}_M(\mathbf{d}, \rho, \mu^{\text{off}}) \times_{(\mathbf{d}, \rho, \mu^{\text{off}})} \mathscr{B}_U(\mathbf{d}, \rho, \mu^{\text{off}}).$$

For each member, define the private institutional contrast using the same unanimity-minus-majority orientation as the public contrast:

$$\Delta V^A(o) = V_U^A(o) - V_M^A(o), \qquad \Delta V_E^A = (1-p)\Delta V^A(\ell) + p\Delta V^A(h).$$

The affine average is applied only after preserving the linked type vector. If either source is empty in the common fiber, the comparison is empty.

The exact correspondence may remain set-valued, but one selection-free region is available. For every nonempty common fiber, if

$$\beta h < \frac{e}{m},$$

then, for both types and ex ante,

$$V_U^A(o) - V_M^A(o) \leq -\beta\left(\frac{e}{m} - \beta h\right) < 0.$$

At equality the same ranking is weak. This condition is sufficient, not necessary. It follows from the lower bound $V_M^A(o) \geq v_M^{\text{safe}}$ and the upper bound $V_U^A(o) \leq v_U^A(h)$; outside it, the full correspondence determines the ranking.

6.3 Public power, informational rents, and type incidence

For each linked private record, define the agenda-game informational rent type by type,

$$IR_g^A(o) = V_g^A(o) - v_g^A(o), \qquad \Delta IR^A(o) = IR_U^A(o) - IR_M^A(o).$$

Because every institutional contrast now uses the orientation unanimity minus majority, the decomposition is

$$\boxed{\Delta V^A(o) = \Delta v^A(o) + \Delta IR^A(o)}, \qquad \Delta IR^A(o) = \Delta V^A(o) - \Delta v^A(o).$$

The same identity holds ex ante after applying the affine map to the same linked vector.

The incidence under unanimity is exact. In every existing fiber, $IR_U^A(\ell) \geq 0$, $IR_U^A(h) \leq 0$, and at least one inequality is strict. Positive informational rent belongs to the low type, which can receive a concession designed for the high type. The high type receives no such premium and may lose relative to its public benchmark. When $\Delta v^A(\ell) < 0$, public information favors majority for the low type. If a comparable private member nevertheless has $\Delta V^A(\ell) > 0$, then member by member

$$\Delta IR^A(\ell) > -\Delta v^A(\ell) > 0.$$

Information is then the only component pointing toward unanimity and more than offsets majority's public advantage. This statement does not select a member or imply a global sign for $\Delta IR^A(\ell)$.

Agenda also changes the informational rent relative to the no-agenda benchmark. Let IR_g^B be the frozen Round-1 correspondence from the baseline. Its date-A value is βIR_g^B, so

$$I_g = IR_g^A - \beta IR_g^B, \qquad \Delta I = \Delta IR^A - \beta \Delta IR^B.$$

This is the only multiplication by β in the interaction. Under unanimity, agenda weakly reduces informational rent in every existing high-prior fiber; under majority and across rules the interaction remains set-valued wherever its sources do.

6.4 The structural effect of the agenda stage

The agenda effect compares presence and absence of the mandatory date-A stage within a fixed information regime. All four arms in Figure 6 are expressed at date A.

	Baseline without hegemonic agenda B	Hegemonic agenda A
Public type	$\beta v_g^B(o)$	$v_g^A(o)$
Private type	$\beta V_g^B(o)$	$V_g^A(o)$

$$\underbrace{T_g(o)}_{\text{agenda under private information}} = \underbrace{D_g(o)}_{\text{agenda under public information}} + \underbrace{I_g(o)}_{\text{agenda–information interaction}}.$$

Figure 6: The factorial agenda-information design. The treatment adds an earlier, mandatory proposal stage; the control begins in Round 1 and is discounted once when evaluated at date A.

Formally,

$$D_g(o) = v_g^A(o) - \beta v_g^B(o), \qquad T_g(o) = V_g^A(o) - \beta V_g^B(o),$$

and for every complete record in which both arms exist,

$$\boxed{T_g(o) = D_g(o) + I_g(o)}.$$

This is a structural causal contrast inside the model, not an empirically identified estimand. Under multiplicity it is a correspondence of admissible contrasts.

The public effects are

$$D_U(o) = 1 - \beta$$

and

$$D_M(o) = \begin{cases} v_M^{\text{safe}} - \dfrac{e}{m}\beta^2 o, & o \le 1/m, \\ v_M^{\text{safe}} - \beta o, & 1/m < o < o_M^*, \\ 0, & o \ge o_M^*. \end{cases}$$

Thus agenda never lowers H's public payoff under majority, but it can have no direct value when the mandatory proposal induces the same delayed continuation as the control. The first branch applies at $o = 1/m$; the frozen continuation selection creates a downward jump in D_M immediately to its right. The threshold o_M^* can lie outside $o \in (0, 1)$.

Under private unanimity, T_U weakly benefits each type wherever both arms exist, with exact cells reported in Appendix E. Under private majority,

$$T_M = D_M + I_M$$

is the exact translation of the interaction correspondence by the fixed public vector. No general sign is imposed. Across rules, member by member within a complete admissible product,

$$\Delta D(o) = D_U(o) - D_M(o), \qquad \Delta T(o) = T_U(o) - T_M(o) = \Delta D(o) + \Delta I(o).$$

The diagonal object $Q_g = v_g^A - \beta V_g^B$ changes agenda and information simultaneously. It is useful for comparison but is not a one-factor causal effect; Appendix E reports it separately.

7 Discussion

7.1 Pivotality, substitutes, and contested strength

The model's central distinction is technological. Under majority, the proposer can replace the hegemon with another uninformed weak-state vote. Under unanimity, the hegemon's approval is an essential input. Removing the substitute changes both bargaining power and the way private information enters proposals.

The complete-information benchmark shows the power component. If the known hegemon is cheap enough, majority includes it and both rules pay the same discounted threshold. If it is expensive, majority excludes it while unanimity must still buy its vote. The informational comparison asks a different question: after netting out that public-type difference, how much does private information change each rule's payoff?

The answer is not simply that private information always benefits a hegemon. Pooling under unanimity benefits the low type because weak states price it as high. Screening under majority can remove that rent, while exclusion can make the comparison depend on whether the high threshold discounted to the present exceeds the low current disagreement payoff. The high type's rent changes only when its treatment differs across the private and public majority games.

The empty middle-belief cell is equally important for scope. It records the failure of every pure voting pattern under the declared belief and pivotality conditions. It does not supply a payoff for institutional comparison and does not justify filling the gap by interpolation.

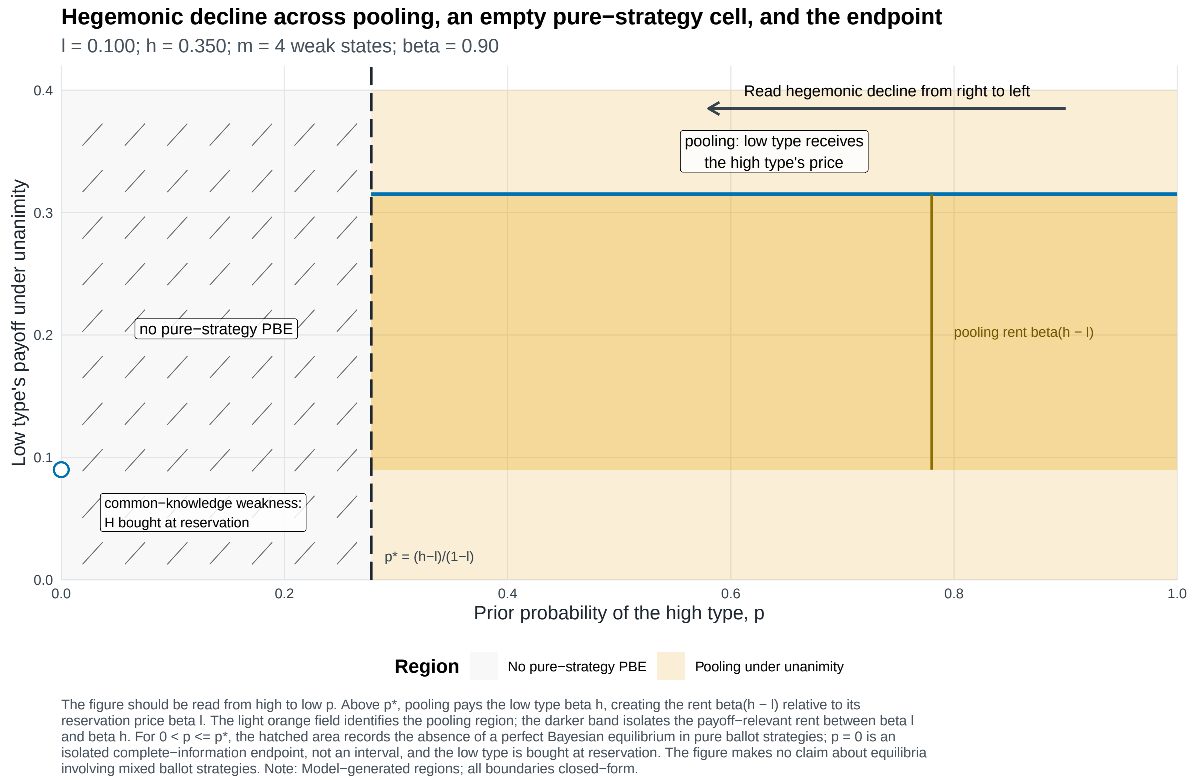

Figure 7: A schematic movement in beliefs about hegemonic strength. The hatched region is the set in which the unanimity game has no equilibrium in pure ballot strategies under the maintained concept. Historical annotations, if used, are illustrations rather than empirical tests or calibration.

7.2 WTO creation, exit, and observable implications

The close of the Uruguay Round provides a sharper institutional illustration than OPEC. Steinberg reports that the United States and the European Community backed the single undertaking and, after joining the WTO, withdrew from GATT 1947 (Steinberg 2002). Their withdrawal terminated the old GATT obligations, including most-favored-nation treatment, toward countries that did not accept the Final Act and join the new organization. The maneuver made the preexisting trading relationship a less valuable fallback and helped give the WTO package broad membership.

The two parts of that strategy map to different bargaining objects. The single undertaking joined the Uruguay Round agreements into one package and restricted selective acceptance. Withdrawal from GATT 1947 changed the value of rejecting that package. The latter supplies a concrete micro-foundation for disagreement payoffs: market power permits a major state to shape what agreement failure is worth to itself and to its partners. If the precise value of the major state's outside option is not common knowledge, bargaining over an approval that cannot be replaced also creates the screening problem analyzed in the model.

The episode is not a literal implementation of the baseline. Steinberg shows that the United States and European Community exercised substantial de facto agenda influence, whereas the benchmark denies the hegemon an exclusive formal proposal right. The benchmark isolates one channel inside a historically richer process. It asks what the consensus requirement contributes after agenda control is held fixed, not whether agenda control was absent or whether the model identifies why the United States created the WTO.

This distinction yields observable implications. Informational power through pivotality should be strongest where a major state's approval has no coalition substitute, its fallback is hard for partners to price, and the negotiated package can vary enough to compensate uncertain participation constraints. In the pooling region, the signature is a smooth agreement that grants a concession suited to a stronger possible type. A theory of signaling through rejection instead predicts visible delay or failed proposals before beliefs change. Under majority, the availability of a substitute coalition should reduce the return on private information even when the underlying outside option remains unequal.

Steinberg also emphasizes information flowing in the opposite direction: the consensus process helps powerful agenda setters elicit weaker states' preferences. The model studies weak proposers who are uncertain about the hegemon's continuation value. An empirical application must distinguish these directions of information and use language of consistency rather than treat the WTO episode as identification of the formal mechanism.

7.3 Limits

The baseline fixes the voting rule, keeps the institutional pie at one, gives proposal power only to weak states, and uses two types and two rounds. The agenda extension relaxes only that baseline proposal restriction by adding the earlier mandatory proposal stage for H. Neither part analyzes endogenous rule choice or an endogenous agenda protocol; the paper also does not analyze more than two types, sequential roll-call voting, or intrinsic agreement benefits. The restriction $m \geq 3$ excludes the three-player case. The strict discounting assumption excludes the corner $\beta = 1$, where additional indifferences would require a separate analysis.

The endpoint-support condition also matters. A type assigned zero probability by the prior cannot be resurrected after an off-path action. This makes the degenerate-prior cells coincide with their complete-information games. In the interior, both types remain in the prior's support and off-path actions by H can sustain a range of posterior beliefs.

The agenda extension preserves rather than resolves equilibrium multiplicity. Its majority effects, institutional comparisons, and some interactions are correspondences. Every robust sign is therefore setwise, and every other sign is member-specific. Empty cells record absence of a pure PBE under M/S/B, not a zero effect. The structural contrast T also depends on the maintained date convention and mandatory proposal stage; it is distinct from the diagonal comparison Q.

Finally, the WTO discussion is a theoretical application. The model neither identifies why the United States and European Community designed the WTO nor establishes that a particular historical actor was a low or high type. It also abstracts from plurilateral agreements, subsidiary voting procedures, ratification, implementation, and repeated-game reputation. Those mechanisms may complement or substitute for the channels isolated here.

8 Conclusion

Consensus can benefit a hegemon even when weak states control the agenda because unanimity removes substitutes for the hegemon's vote. Complete information reveals the familiar pivotality component. Private information adds a distinct component: when unanimity pools at the high threshold, a low type can earn rent; whether this exceeds its majority rent depends on the majority correspondence and the outside-option region.

Restoring hegemonic agenda power does not erase the benchmark result. It separates real power from its informational shadow. When the outside option is high, the public component can itself favor unanimity. When public information favors majority, a private unanimity ranking can arise only if the relative informational rent offsets that public disadvantage. The structural effect of the agenda stage is $T = D + I$: its direct value and its interaction with information are distinct, and neither has a universal institutional sign.

Formal equality of votes can therefore coexist with unequal bargaining power through three channels: outside options, proposal rights, and the price of an approval with no substitute. The WTO application is consistent with their coexistence, while the identifying benchmark shows that the last channel does not require an exclusive formal proposal right. The paper's conclusions remain conditional on complete linked correspondences, their domains, and the cells in which the maintained equilibrium concept exists.

Appendix A: Protocol, beliefs, and incentives

A.1 Complete transition and payoff rules

For a proposal $x = (x_H, x_1, \dots, x_m)$, let $a_H \in \{Y, N\}$ and let the weak responders' votes be a_{-i}. The proposer votes yes. If the quota is met, the proposal is implemented in full. Each weak respondent receives x_j, the proposer receives x_i, and H receives x_H after yes and $x_H + o$ after no. If the quota is not met in Round 1, the public vote vector leads to Round 2; every continuation payoff is then multiplied by β. If the quota is not met in Round 2, weak states receive zero and H receives o.

The $x_H + o$ branch is necessary for a complete strategy specification. It is not an equilibrium description. Whenever majority passes without H, a proposer strictly prefers to move any positive x_H into its own allocation, holding all votes fixed. Hence all equilibrium exclusions set $x_H = 0$ and pay H exactly o.

A.2 Beliefs and ballot restrictions

Uninformed weak-state proposals and votes preserve the current belief. A vote by H updates by Bayes' rule when its denominator is positive. At a zero-denominator action in an interior-prior game, the posterior may range over $[0, 1]$, subject to structural consistency across histories that encode the same information. At $p = 0$ or $p = 1$, posterior support never expands.

A weak responder compares its allocation with the continuation that would follow if its vote switched passage into failure. It votes yes at equality. H maximizes its type-contingent continuation payoff and votes yes at exact indifference. These restrictions apply at every proposal, including off-path proposals.

Appendix B: Proofs

B.1 Proof of Proposition 5.1

In terminal majority, a weak responder's disagreement value is zero, so any nonnegative allocation induces yes under the indifference-to-yes convention. Because $k \leq m$, the proposer can pass without H, set every payment to zero, and keep one. H is nonpivotal and strictly votes no because no yields $x_H + o$ rather than x_H. Moving any positive x_H to the proposer leaves passage unchanged, so equilibrium exclusion has $x_H = 0$ and H receives o.

In terminal unanimity, every weak responder accepts zero and H accepts exactly when $x_H \geq o$. The proposer therefore sets $x_H = o$.

Move to Round 1. The terminal continuation of a weak state before recognition under majority is $1/m$, hence $w = \beta/m$. The continuation of H is βo. Under unanimity, paying H exactly βo and each weak responder its continuation implements immediate agreement and leaves the proposer strictly more than delay because $\beta < 1$.

Under majority, inclusion costs $(k-1)w + \beta o$; exclusion costs kw. Inclusion is weakly cheaper exactly when $o \leq 1/m$. At equality the proposer is indifferent, and the proposal tie-break selects inclusion because H receives $\beta o < o$. This gives the stated payoffs. □

B.2 Proof of Proposition 5.2

The majority argument is the terminal-majority part of the preceding proof. Before recognition, symmetry gives each weak state expected payoff $1/m$.

Under unanimity, weak responders accept zero. Any accepted offer to H can be reduced to the lowest threshold consistent with its acceptance set. The only nonempty acceptance sets are the low type alone, implemented by $x_H = \ell$, and both types, implemented by $x_H = h$. The first gives the proposer $(1-p)(1-\ell)$; the second gives $1-h$. The low offer is weakly preferred exactly when

$$p \leq \frac{h-\ell}{1-\ell} = p^*.$$

At equality the low offer gives H the lower expected payoff and is selected. □

B.3 Proof of Proposition 5.3

A responding weak state votes yes exactly when $x_j \geq w = \beta/m$. Let n_Y be the number of such responders. If $n_Y \geq k$, the weak votes pass the proposal without H; both types of H vote no and receive $x_H + o$. If $n_Y = k - 1$, H is pivotal and a type with outside option o votes yes exactly when $x_H \geq \beta o$. If $n_Y \leq k - 2$, the proposal fails regardless of H's vote.

Within each acceptance class, reducing weak payments to w, reducing x_H to the relevant type threshold, and assigning the remaining pie to the proposer weakly increases its payoff. This leaves four candidates: exclusion with payoff Π_E, screening with payoff $\Pi_S(p)$, pooling with payoff Π_P, and deliberate delay with payoff w. Exclusion is always feasible and

$$\Pi_E - w = 1 - \frac{\beta(k+1)}{m} > 0,$$

so delay is never selected. A screening or pooling candidate that could beat exclusion is automatically feasible.

The decisive payoff differences are

$$\Pi_P - \Pi_E = \beta(1/m - h)$$

and

$$\Pi_S(p) - \Pi_E = (1-p)\beta(1/m - \ell) - p(1 - \beta(k+1)/m).$$

Where pooling can beat exclusion, comparing screening with pooling yields $p_{S=P}$; comparing screening with exclusion yields $p_{S=E}$. Substituting the signs of $\ell - 1/m$ and $h - 1/m$ produces the five cases in Proposition 5.3.

At screening ties, its expected payoff to H is strictly smaller than that of the tied alternative, so the proposal tie-break selects screening. When $h = 1/m$, exclusion and pooling give the proposer the same payoff. Their payoffs to H determine the stated selection; exact equality leaves their entire proposal segment. All selected proposals exhaust the pie. Finally, permuting the identities of weak responders does not change any comparison, which establishes the reported identity multiplicity. $\square$

B.4 Proof of Proposition 5.4

Transport the terminal unanimity continuation into Round-1 units. If the posterior is μ, a weak state's continuation is

$$W(\mu) = \begin{cases} (1-\mu)w_\ell^U, & \mu \le p^*, \\ w_h^U, & \mu > p^*. \end{cases}$$

Thus $W(\mu) \in [w_h^U, w_\ell^U]$, and $x_j = w_\ell^U$ forces every weak responder to vote yes under any admissible posterior.

At $p = 0$, support preservation fixes every posterior at zero. The proposal

$$x_H = \beta\ell, \qquad x_j = w_\ell^U, \qquad x_i = 1 - \beta\ell - (m-1)w_\ell^U = w_\ell^U + 1 - \beta$$

is accepted. Each responder and H is exactly indifferent and votes yes. Any lower payment loses a required vote; any higher payment reduces the proposer's residual. A delay gives the proposer w_ℓ^U, so immediate agreement is strictly better by $1-\beta$.

For $p > p^*$, the analogous pooling proposal is

$$x_H = \beta h, \qquad x_j = w_h^U, \qquad x_i = 1 - \beta h - (m-1)w_h^U = w_h^U + 1 - \beta.$$

Both types of H and every weak responder accept, and immediate agreement again exceeds delay by $1-\beta$.

For completeness, write $u = \min_j x_j$ after an arbitrary proposal and order the strategy of H as low-type/high-type. At $p = 0$, posterior support is always the low type. The profile (Y, Y) is sequentially rational if $u < w_\ell^U$, or if $u \ge w_\ell^U$ and $x_H \ge \beta h$; (N, N) is rational if $u \ge w_\ell^U$ and $x_H < \beta\ell$; and (Y, N) is rational if $u \ge w_\ell^U$ and $\beta\ell \le x_H < \beta h$. The profile (N, Y) is never rational. These cases are mutually exclusive and exhaustive.

For $p^* < p < 1$, the profile (Y, Y) is rational if $u < w_h^U$, or if $u \ge w_h^U$ and $x_H \ge \beta h$; (N, N) is rational if $u \ge w_h^U$ and $x_H < \beta h$; and neither separating profile is rational. Bayes gives the current prior after an on-profile pooled action. When yes is a zero-probability action, an admissible posterior μ_Y supports all weak yes votes exactly when $W(\mu_Y) \le u$, which is possible in the stated no profile because $u \geq w_h^U$. At $p = 1$, support preservation fixes the posterior at one and the same two pooled-profile conditions apply. Thus every proposal in both existence domains has a pure, sequentially rational response completion.

It remains to show the empty cell. For $0 < p \leq p^*$, consider

$$s^\dagger : \quad x_H = \beta\ell, \quad x_j = w_\ell^U \text{ for every weak responder}, \quad x_i = 1 - \beta\ell - (m-1)w_\ell^U.$$

All weak responders must vote yes. Enumerate the four pure type-contingent profiles for H, ordered low/high:

1. Under (Y, Y), the high type receives $\beta\ell < \beta h$ and profitably votes no to obtain continuation βh.
2. Under (N, N), the low type's no gives $\beta\ell$, so the indifference-to-yes convention requires yes.
3. Under (Y, N), the low type can imitate the high type's no. That action reveals the high type and yields continuation $\beta h > \beta\ell$.
4. Under (N, Y), the high type can imitate the low type's no and obtain continuation βh, while the prescribed comparison cannot support strict no against yes at the offer.

No pure profile is sequentially rational after $s^\dagger$. Because a PBE must specify a valid pure response after every feasible proposal, no pure-ballot PBE exists in this cell. This also explains the discontinuity: at $p = 0$, support preservation rules out the posterior that drives the imitation argument; for every positive p, both types are possible. $\square$

B.5 Proof of Proposition 5.5

The private majority payoff vectors for H are

$$V_M^{B,S} = (\beta\ell, \beta h), \quad V_M^{B,P} = (\beta h, \beta h), \quad V_M^{B,E} = (\ell, h).$$

The private unanimity vectors are $(\beta\ell, \beta h)$ at $p = 0$, empty for $0 < p \leq p^*$, and $(\beta h, \beta h)$ for $p > p^*$. Componentwise subtraction gives the proposition. On any proposal segment, subtraction is affine in the common proposal weight, so the payoff set and outcome set remain atomic. □

B.6 Proofs of Propositions 5.6 and 5.7

From Proposition 5.1, the public payoff vectors are

$$v_M^B = (v_M^B(\ell), v_M^B(h)), \qquad v_U^B = (\beta\ell, \beta h).$$

Subtracting these from each private vector gives the table in Proposition 5.6. In particular, unanimity is payoff-equivalent to its public endpoint at $p = 0$, is empty in the middle cell, and pools above the cutoff, which yields $(\beta(h - \ell), 0)$.

Subtract IR_M^B from IR_U^B. In the high-belief cell, screening gives $(\beta(h-\ell), 0)$ when both types are publicly included and $(\beta(h - \ell), (1 - \beta)h)$ when only the low type is publicly included; pooling gives zero. Exclusion gives $(\beta h - \ell, -(1-\beta)h)$, $(\beta h - \ell, 0)$, or $(\beta(h - \ell), 0)$, respectively when both types are included, only the low type is included, or both types are excluded. The endpoint follows by the same calculation. An empty source set remains empty. For the residual segment, affine subtraction with its common weight λ yields the one-dimensional segment stated in the propositions, not the Cartesian product of marginal envelopes. □

Appendix C: Endpoints, exact sets, and illustration

C.1 Endpoint equivalence

At $p = 0$, posterior support is the singleton low type. The private unanimity proposal and payoff coincide with the public low-type game. At $p = 1$, posterior support is the singleton high type, and the private pooling proposal and payoff coincide with the public high-type game. The same equivalence holds under majority: applying the relevant private class at each endpoint reproduces public inclusion or exclusion, including the $o = 1/m$ tie-break. Thus the endpoints are literal complete-information games, not one-sided limits.

C.2 Exact correspondence and envelopes

Most parameter cells are singleton payoff vectors. At $h = 1/m$, exact indifference between exclusion and pooling can leave a proposal segment. If λ is the exclusion weight, every reported object uses that same λ. The lower and upper envelope of any component may summarize the segment, but coordinated marginal envelopes do not imply that all pairwise combinations are attainable.

For majority-rule informational rents, the exact componentwise envelopes are

$$IR_M^B(\ell) \in [\min\{(1-\beta)\ell, \beta(h-\ell)\}, \max\{(1-\beta)\ell, \beta(h-\ell)\}], \qquad IR_M^B(h) \in [0, (1-\beta)h].$$

For both the private institutional payoff contrast and ΔIR^B, they are

$$\text{low type} \in [\min\{0, \beta h - \ell\}, \max\{0, \beta h - \ell\}], \qquad \text{high type} \in [-(1-\beta)h, 0].$$

In each case these are the marginal envelopes of a single line segment. The same λ indexes both coordinates, so the Cartesian product of the two intervals is not attainable.

Where $0 < p \le p^*$, the private unanimity correspondence is empty. Accordingly, V_U^B, IR_U^B, and ΔIR^B are empty. No payoff, sign, institutional ranking, or interpolation is assigned to that cell.

C.3 Worked values

Return to $m = 4$, $k = 2$, $\beta = 0.9$, $\ell = 0.10$, and $h = 0.35$. Besides $p^* = 0.2778$, the majority screening–exclusion cutoff is

$$p_{S=E} = \frac{0.9(0.25 - 0.10)}{0.9(0.25 - 0.10) + 1 - 0.9(3/4)} = 0.2935.$$

At $p = 0.80$, majority excludes and unanimity pools. The public payoff vectors are

$$v_M^B = (0.09, 0.35), \qquad v_U^B = (0.09, 0.315),$$

while the private payoff vectors are

$$V_M^B = (0.10, 0.35), \qquad V_U^B = (0.315, 0.315).$$

Therefore

$$IR_M^B = (0.01, 0), \qquad IR_U^B = (0.225, 0), \qquad \Delta IR^B = (0.215, 0).$$

These numbers illustrate one point where the low type is publicly included and the high type excluded; they do not average over parameter values or estimate a real institution.

Appendix D: Notation

Table 7: Notation for the essential-input bargaining game

Symbol	Meaning
H	Privately informed hegemon
W, m	Set and number of weak states; $m \geq 3$
$k = \lfloor (m+1)/2 \rfloor$	Additional yes votes a proposer needs under majority
$o \in \{\ell, h\}$	Hegemon's terminal disagreement payoff, with $0 < \ell < h < 1$
p	Prior probability of the high outside option h
μ, μ^{off}	Posterior probability of h, on and off path
β	One-round discount factor, strictly between zero and one
$x = (x_H, x_1, \dots, x_m)$	Feasible allocation vector
$\bar{x}_H$	Maximum feasible allocation to H; $h \leq \bar{x}_H \leq 1$
x_H, x_j, x_i	Allocations to H, weak responder j, and proposer i
$w = \beta/m$	First-round continuation value of a weak state under majority
p^*	Terminal unanimity cutoff, $(h - \ell)/(1 - \ell)$
B, A	Games without and with the earlier hegemonic agenda stage
$v_g^B(o)$, $v_g^A(o)$	Public-information payoff under rule g in each arm

Symbol	Meaning
$V_g^B(o)$, $V_g^A(o)$	Private-information payoff correspondence in each arm
IR_g^B, IR_g^A	Private payoff minus the same-arm public benchmark
ΔIR^B, ΔIR^A	Unanimity-minus-majority informational-rent contrasts
$e = m - k$	Weak states excluded by a minimal majority coalition
v_M^{safe}	Safe majority-agenda payoff, $1 - k\beta/m$
o_M^*	Majority-agenda delay cutoff, v_M^{safe}/β
ρ	Off-path likelihood-ratio coordinate in the agenda extension
$\Delta v^A(o)$	Public institutional gap with agenda, $v_U^A(o) - v_M^A(o)$
$\Delta V^A(o)$	Private institutional contrast with agenda, $V_U^A(o) - V_M^A(o)$
D_g, I_g, T_g	Direct public agenda effect, agenda–information interaction, and total private agenda effect
Q_g	Diagonal agenda-only versus information-only comparison

Appendix E: Exact agenda-stage correspondences and comparisons

E.1 Agenda-stage contract

Fix

$$\mathbf{d} = (m, k, e, \beta, \ell, h, \bar{x}_H, \mathcal{X}, p),$$

$$m \geq 3, \quad k = \lfloor (m+1)/2 \rfloor, \quad e = m - k,$$

$$0 < \beta < 1, \quad 0 < \ell < h < 1, \quad h \leq \bar{x}_H \leq 1, \quad p \in [0, 1].$$

At date A, after observing its type, H proposes

$$x^A = (x_H, x_1, \dots, x_m) \in \mathcal{X}, \qquad \mathcal{X} = \{x^A : x_H \geq 0,\ x_j \geq 0,\ x_H + \textstyle\sum_j x_j \leq 1\}.$$

The proposal counts as H's affirmative vote. Weak states vote simultaneously and in pure strategies. Majority requires k affirmative weak votes; unanimity requires all m. Passage implements the package at date A. Rejection enters the corresponding frozen Round-1 continuation, whose native payoff is multiplied by β exactly once.

A rejected public history includes the rule, proposal, proposer, full ballot, outcome, and posterior. Its continuation selector is total, public, Borel, common to types compatible with the history, and returns one complete frozen Round-1 equilibrium record. A record binds strategies, beliefs, recognition lotteries, ballots, payoffs, and outcomes. No scalar continuation payoff may replace it.

For an interior prior, the equilibrium proposal laws of the two types are Borel measures σ_ℓ, σ_h. At every disciplined proposal, Bayes' rule determines the posterior from their local likelihood ratio. At undisciplined proposals, the M/S/B restriction fixes the posterior to

$$\mu^{\text{off}} = b_\rho(p) = \frac{p\rho}{1 - p + p\rho}, \qquad \rho \in [0, \infty].$$

The extended-real boundary conventions are

$$b_0(p) = 0, \qquad b_\infty(p) = 1$$

for an interior prior. The endpoints preserve prior support: $\mu^{\text{off}} = p$ everywhere, the fiber is $(*, p)$, and ρ is irrelevant. Voting follows the same as-if-pivotal comparison and indifference-to-yes convention as the baseline.

E.2 Complete private-majority correspondence

For each posterior μ and a fixed anonymous Markov continuation selection χ, let $A_\chi(\mu)$ be H's best payoff from a proposal that passes and let $D_{\chi,o}(\mu)$ be the payoff of a type with outside option o from rejection, both at date A. Define

$$A_\mu := A_\chi(\mu), \qquad D_{o,\mu} := D_{\chi,o}(\mu), \qquad \mu \in [0, 1],$$

so that A_p, A_0, A_1 and $D_{\ell,p}, D_{h,p}, D_{\ell,0}, D_{\ell,1}, D_{h,0}, D_{h,1}$ denote evaluations of these same functions at the displayed posterior. Also define

$$O_o(\rho) = \max\{A_\chi(b_\rho(p)), D_{\chi,o}(b_\rho(p))\}.$$

The following conditions are necessary and sufficient for every interior pure equilibrium class:

Table 8: Pure private-majority classes at the agenda stage

Class	Necessary and sufficient condition and linked type payoff
Pooling agreement	$O_h(\rho) \le A_p$, with any $z \in [O_h(\rho), A_p]$; linked payoff (z, z)
Pooling delay	$D_{\ell,p} \ge O_\ell(\rho)$ and $D_{h,p} \ge O_h(\rho)$; linked payoff $(D_{\ell,p}, D_{h,p})$
Separating agreement–agreement	$O_h(\rho) \le \min\{A_0, A_1\}$, with any $z \in [O_h(\rho), \min\{A_0, A_1\}]$; linked payoff (z, z)
Low agreement, high delay	$D_{h,1} \ge O_h(\rho)$ and $\max\{D_{\ell,1}, O_\ell(\rho)\} \le \min\{A_0, D_{h,1}\}$; linked payoff $$(z, D_{h,1}), \qquad z \in [\max\{D_{\ell,1}, O_\ell(\rho)\}, \min\{A_0, D_{h,1}\}]$$
Low delay, high agreement	Impossible; linked payoff $\emptyset$
Separating delay–delay	$D_{\ell,0} \ge D_{\ell,1}$, $D_{h,1} \ge D_{h,0}$, $D_{\ell,0} \ge O_\ell(\rho)$, and $D_{h,1} \ge O_h(\rho)$; linked payoff $(D_{\ell,0}, D_{h,1})$

These classes preserve proposals and messages even when their payoffs coincide. Separating agreement at the same z is separation in messages at indifference, not pooling. The off-path feasibility set for ρ can be disconnected because the frozen majority continuation changes branches; no single cutoff substitutes for the table.

For mixed strategies, a complete reduced record is

$$R_M = (\rho, \mu^{\text{off}}, \sigma_\ell, \sigma_h, \lambda, \pi, \chi, a, u_\ell, u_h).$$

It retains the public proposal law λ, posterior map π, continuation selection χ, ballot map a, and both type payoffs. Bayes holds pointwise on the disciplined support, every weak vote satisfies the pivotal comparison at every proposal, and each type's proposal law is supported on its global best responses. This criterion is necessary and sufficient; arbitrary Borel mixtures and atomless supports are included.

At $p \in \{0, 1\}$, the posterior is fixed at the endpoint. With

$$M_o = \max\{A_p, D_{o,p}\},$$

each type's Borel proposal law may be any probability measure supported on its argmax. The probability-zero type remains part of the linked record. For every admissible primitive and prior, at least one majority PBE exists for some ρ; this does not assert existence at every fixed ρ.

E.3 Complete private-unanimity correspondence

Define

$$r_U(o) = \frac{\beta(1-\beta o)}{m}, \qquad v_U^A(o) = 1 - \beta + \beta^2 o.$$

Thus $0 < r_U(h) < r_U(\ell)$, and the two proposal primitives are

$$x^\ell = (v_U^A(\ell), r_U(\ell), \dots, r_U(\ell)), \qquad x^h = (v_U^A(h), r_U(h), \dots, r_U(h)).$$

At posterior zero, every weak state accepts exactly when $x_j \geq r_U(\ell)$; at a high posterior, it accepts exactly when $x_j \geq r_U(h)$. Equality passes under the indifference-to-yes convention. The only continuation posteriors admitted by the frozen unanimity continuation are

$$\mathcal{D}_C = \{0\} \cup (p^*, 1].$$

Rejection pays the high type $\beta^2 h$ at every admissible posterior; it pays the low type $\beta^2 \ell$ at posterior zero and $\beta^2 h$ at a high posterior.

A complete unanimity member is the indivisible binder

$$R_U = (\sigma_\ell, \sigma_h, \mu, \mu^{\text{off}}, \hat{\kappa}_U, v, \Omega, \Gamma_\ell, \Gamma_h).$$

Here σ_o is a Borel proposal law, $\bar{\sigma} = (1-p)\sigma_\ell + p\sigma_h$, μ is the public posterior, and $\mu^{\text{off}} \in \mathcal{D}_C$ is its single value at every undisciplined proposal. The selector $\hat{\kappa}_U$ returns a complete literal frozen continuation member at the same posterior, v is the full pure ballot map, and $\Omega_o(x)$ is the literal terminal law induced by passage or by that continuation. Finally, Γ_o is the realized pushforward law of the joint record

$$(x, \mu(x), v(x), \text{pass/reject}, \hat{\kappa}_U, \Omega_o(x), \text{date-}A \text{ payoffs}).$$

The local Bayes limit must exist at every disciplined proposal, including zero-mass limit points, and must lie in $\mathcal{D}_C$. At each such point, and at every undisciplined point evaluated with μ^{off}, neither type may obtain a profitable proposal deviation. These pointwise restrictions, the ballot map, the literal continuation, and both realized laws are part of the same binder and cannot be recombined across members.

The full interior correspondence is generated by the following two families.

Low-posterior family AU-MSB-L. Its exact domain is

$$0 < p < 1, \qquad v_U^A(\ell) \geq \beta^2 h, \qquad \mu^{\text{off}} = 0.$$

Choose Borel laws satisfying all of the following:

1. $\sigma_\ell(\{x^\ell\}) = \alpha_L > 0$ and $\sigma_h(\{x^\ell\}) = 0$;
2. outside x^ℓ, $\mu(x) > p^*$ $\bar{\sigma}$-almost surely;
3. each used proposal outside x^ℓ either passes with H's share $v_U^A(\ell)$, or, only when $v_U^A(\ell) = \beta^2 h$, rejects and pays $\beta^2 h = v_U^A(\ell)$;
4. every disciplined proposal, including every zero-mass limit point, gives each type a deviation

payoff no larger than $v_U^A(\ell)$; every undisciplined proposal satisfies the same inequality under posterior zero.

Every accepted high-posterior signal therefore belongs to

$$\mathcal{K}_L = \{x \in \mathcal{X} : x_H = v_U^A(\ell),\ \min_j x_j \geq r_U(h)\}.$$

The laws may be asymmetric, discrete, semi-pooling, mixed, or atomless. A pure witness has the low type propose x^ℓ and the high type propose $x^S = (v_U^A(\ell), r_U(h), \dots, r_U(h))$. Every member has linked payoff $(v_U^A(\ell), v_U^A(\ell))$. If $v_U^A(\ell) > \beta^2 h$, agreement occurs with probability one for both types; if $v_U^A(\ell) = \beta^2 h$, high-posterior rejected signals may carry mass. The weak-state payoff attached to the same member is x_j after passage, $r_U(\ell)$ after rejection at $(\mu, o) = (0, \ell)$, zero after rejection at $(0, h)$, and $r_U(h)$ after rejection at any high posterior.

High-posterior family AU-MSB-H. Its necessary prior domain is $p^* < p < 1$. When $\mu^{\text{off}} = 0$, choose any

$$u \in [\max\{v_U^A(\ell), \beta^2 h\}, v_U^A(h)],$$

and Borel laws such that: $\mu > p^*$ $\bar{\sigma}$-almost surely; the local admissible Bayes limit exists at every disciplined point; every used accepted proposal lies in

$$\mathcal{K}_H(u) = \{x \in \mathcal{X} : x_H = u,\ \min_j x_j \geq r_U(h)\};$$

used rejected proposals occur only when $u = \beta^2 h$; every disciplined deviation gives each type at most u; and every undisciplined deviation, evaluated at posterior zero, also gives at most u. The linked payoff is (u, u). If $u > \beta^2 h$, agreement has probability one for each type; if $u = \beta^2 h$, each type may mix accepted proposals with $x_H = \beta^2 h$ and rejected high-posterior proposals. A pure pooling witness for every u is

$$x(u) = \left(u, \frac{1-u}{m}, \dots, \frac{1-u}{m}\right), \qquad \sigma_\ell = \sigma_h = \delta_{x(u)}, \qquad \mu(x(u)) = p.$$

When $\mu^{\text{off}} \in (p^*, 1]$, the only current proposal laws are $\sigma_\ell = \sigma_h = \delta_{x^h}$, with $\mu(x^h) = p$ and payoff $(v_U^A(h), v_U^A(h))$; multiplicity survives only in μ^{off} and the complete literal continuation binder.

When $\mu^{\text{off}} \in (0, p^*]$, the correspondence is empty. For $p^* < p < 1$ and $\mu^{\text{off}} = 0$, the high-posterior family always exists, and the low-posterior family is additionally present exactly when $v_U^A(\ell) \geq \beta^2 h$. For $0 < p \leq p^*$, equilibrium exists exactly when $v_U^A(\ell) \geq \beta^2 h$ and consists only of the low-posterior family. There is no third interior family.

The endpoints are complete Borel strategy correspondences, not one-sided limits. At $p = 0$, support preservation fixes $\mu(x) = \mu^{\text{off}} = 0$ everywhere, $\sigma_\ell = \delta_{x^\ell}$, and the counterfactual high-type law is

$$\begin{cases} \delta_{x^\ell}, & v_U^A(\ell) > \beta^2 h, \\ \text{any Borel law supported on } \{x^\ell\} \cup \mathcal{R}_L, & v_U^A(\ell) = \beta^2 h, \\ \text{any Borel law supported on } \mathcal{R}_L, & v_U^A(\ell) < \beta^2 h, \end{cases}$$

where

$$\mathcal{R}_L = \{x \in \mathcal{X} : \min_j x_j < r_U(\ell)\}.$$

The linked payoff is $(v_U^A(\ell), \max\{v_U^A(\ell), \beta^2 h\})$, and the low type agrees immediately with probability one. At $p = 1$, support preservation fixes $\mu(x) = \mu^{\text{off}} = 1$ everywhere and $\sigma_\ell = \sigma_h = \delta_{x^h}$, including the probability-zero type's strategy, with linked payoff

$$V_U^A = (v_U^A(h), v_U^A(h)).$$

Consequently, the payoff image of the complete correspondence is

$$V_U^A = \begin{cases} (v_U^A(\ell), \max\{v_U^A(\ell), \beta^2 h\}), & p = 0, \\ (v_U^A(\ell), v_U^A(\ell)), & 0 < p \leq p^*,\ v_U^A(\ell) \geq \beta^2 h,\ \mu^{\text{off}} = 0, \\ \{(u, u) : u \in [\max\{v_U^A(\ell), \beta^2 h\}, v_U^A(h)]\}, & p^* < p < 1,\ \mu^{\text{off}} = 0, \\ (v_U^A(h), v_U^A(h)), & p^* < p < 1,\ \mu^{\text{off}} \in (p^*, 1], \\ (v_U^A(h), v_U^A(h)), & p = 1, \\ \emptyset, & \text{all remaining fibers.} \end{cases}$$

This display is only an image: it neither replaces the binder nor converts an interval of linked vectors into a Cartesian product.

For the two-layer representation, let Z_U be the compact realized-record space containing the named proposal and ballot coordinates, the realized posterior, pass/reject, the literal continuation-cell identity, and the terminal allocation and payoff record. Then

$$\Gamma_o^{U,R_U} = \mathrm{Law}_o(x, \mu(x), v(x), \text{pass/reject}, \zeta_U, \omega_{\text{term}}) \in \mathcal{P}(Z_U).$$

Under a common permutation g of weak-state names, let $\mathcal{R}_g$ relabel every named weak-state coordinate and let Λ_γ code the diagonal orbit of $\gamma = (\Gamma_\ell, \Gamma_h)$. For interior priors,

$$\mathrm{Sig}_U^{ex}(R_U) = (\rho, \mu^{\text{off}}, \Lambda_\gamma), \qquad \mathrm{Sum}_U^{econ}(R_U) = (\rho, \mu^{\text{off}}, (q_U)_{\#}\Gamma_\ell, (q_U)_{\#}\Gamma_h),$$

with endpoint forms $(*, p, \Lambda_\gamma)$ and $(*, p, (q_U)_{\#}\Gamma_\ell, (q_U)_{\#}\Gamma_h)$. The exact signature preserves the diagonal orbit of realized laws and their relevant realized identities; the economic layer removes names record by record. Neither layer contains the full off-path functions. Formal-identity and realized-law operations use the exact signature; any operation that depends on off-path beliefs, strategies, supports, messages, or continuation plans must instead consume the underlying complete binder. No Reynolds average is treated as an equilibrium representative, and every `none` cell above remains empty in both layers.

E.4 Exact same-fiber institutional comparison

Let $\mathcal{B}_M(\mathbf{d}, \rho, \mu^{\text{off}})$ and $\mathcal{B}_U(\mathbf{d}, \rho, \mu^{\text{off}})$ be the full correspondences just described. The admissible comparison set is

$$\mathcal{J}_A^{\text{bind}}(\mathbf{d}, \rho, \mu^{\text{off}}) = \mathcal{B}_M(\mathbf{d}, \rho, \mu^{\text{off}}) \times_{(\mathbf{d}, \rho, \mu^{\text{off}})} \mathcal{B}_U(\mathbf{d}, \rho, \mu^{\text{off}}).$$

This product requires identical primitives and the same fiber, but it imposes no common proposal, continuation realization, or cross-world randomization. The exact linked type-payoff contrast is

$$\mathcal{C}_{\ell h}^A(\mathbf{d}, \rho, \mu^{\text{off}}) = \{u - v : u \in V_U^A(\mathbf{d}, \rho, \mu^{\text{off}}),\ v \in V_M^A(\mathbf{d}, \rho, \mu^{\text{off}})\}.$$

It is a difference of complete vectors. The ex ante image is

$$\mathcal{C}_E^A(\mathbf{d}, \rho, \mu^{\text{off}}) = \{(1-p)x_\ell + px_h : (x_\ell, x_h) \in \mathcal{C}_{\ell h}^A(\mathbf{d}, \rho, \mu^{\text{off}})\}.$$

It is generally not equal to the sum of independently chosen marginal images. Dominance claims are setwise: unanimity dominates in a coordinate only when every member has positive sign; majority dominates only when every member has negative sign. A crossing set is selection dependent. An empty fiber has classification `none`, not sign zero.

E.5 The sufficient majority-advantage region

Every majority binder satisfies

$$V_M^A(o) \geq \max\{v_M^{\text{safe}}, \beta^2 o\}, \qquad v_M^{\text{safe}} = 1 - \frac{k\beta}{m},$$

while every unanimity binder satisfies

$$V_U^A(o) \leq v_U^A(h) = 1 - \beta + \beta^2 h.$$

Therefore

$$v_M^{\text{safe}} - v_U^A(h) = \beta\left(\frac{e}{m} - \beta h\right).$$

If $\beta h < e/m$, then this margin is positive and every comparable member satisfies

$$V_U^A(o) - V_M^A(o) \leq -\beta\left(\frac{e}{m} - \beta h\right) < 0$$

for both types and ex ante. Equality yields weak majority advantage. This proves sufficiency only.

The condition is not necessary. With

$$m = 4,\ k = 2,\ e = 2,\ \beta = .9, \ell = .5,\ h = .6,\ \bar{x}_H = .8,\ p = 0,$$

we have $e/m = .5$ and $\beta h = .54 > .5$, so the condition fails. Nevertheless $v_M^{\text{safe}} = .55$, $v_U^A(\ell) = .505$, and $\beta^2 h = .486$; hence every majority binder strictly exceeds the endpoint unanimity vector

$(.505, .505)$.

The excluded fraction is

$$\frac{e}{m} = \begin{cases} 1/2, & m \text{ even}, \\ (m-1)/(2m), & m \text{ odd}. \end{cases}$$

In an existing low-prior cell, the weaker local condition $\beta\ell < e/m$ gives the margin $\beta(e/m - \beta\ell)$. At $p = 0$, that condition guarantees only the ex ante conclusion unless the full counterfactual type vector is used.

E.6 Public-majority PBE classes

For public $o \leq 1/m$, each weak vote costs

$$r_M(o) = \frac{\beta(1-\beta o)}{m}.$$

Every optimal proposal pays exactly $r_M(o)$ to exactly k weak states, pays zero to the rest, and leaves

$$v_M^A(o) = 1 - \frac{k\beta(1-\beta o)}{m}$$

to H. Every probability measure over these minimal coalitions is a PBE proposal lottery. Passage strictly dominates the delayed payoff $\beta^2 o$.

For $o > 1/m$, every purchased weak vote costs β/m, so passage leaves v_M^{safe} and rejection yields βo. Thus:

$$\begin{aligned} 1/m < o < o_M^* \quad &: \quad \text{every PBE passes through a minimal coalition,} \\ o = o_M^* \quad &: \quad \text{every linked mixture of minimal passage and delay is allowed,} \\ o > o_M^* \quad &: \quad \text{every PBE deliberately delays.} \end{aligned}$$

At the tie, the same mixture weight binds the proposal, result, and agreement probability. The public payoff is always the singleton

$$v_M^A(o) = \begin{cases} 1 - \dfrac{k\beta(1-\beta o)}{m}, & o \le 1/m, \\ \max\{v_M^{\text{safe}}, \beta o\}, & o > 1/m. \end{cases}$$

E.7 Public-unanimity PBE

Every weak vote costs

$$r_U(o) = \frac{\beta(1-\beta o)}{m}.$$

The unique optimal on-path proposal pays this amount to all m weak states and leaves

$$v_U^A(o) = 1 - \beta + \beta^2 o$$

to H. Rejection yields $\beta^2 o$, so passage is strictly better by $1-\beta$. Agreement is immediate and payoff-determined for every public type; no majority-style delay is imported into this rule.

E.8 The public gap

Subtracting E.6 from E.7 gives the unanimity-minus-majority gap

$$\Delta v^A(o) = \begin{cases} -\beta(e/m)(1-\beta o), & o \le 1/m, \\ \beta(\beta o - e/m), & o > 1/m,\ \beta o \le v_M^{\text{safe}}, \\ (1-\beta)(1-\beta o), & o > 1/m,\ \beta o \ge v_M^{\text{safe}}. \end{cases}$$

Continuity holds at $\beta o = v_M^{\text{safe}}$. For $o > 1/m$,

$$\text{sgn}\, \Delta v^A(o) = \text{sgn}(\beta o - e/m).$$

At $o = 1/m$, the first branch applies because the frozen no-agenda continuation includes H at equality. This special selection also matters for the direct agenda effect in E.12.

E.9 Exact informational-rent translations

For every complete private binder,

$$IR_g^A = V_g^A - (v_g^A(\ell), v_g^A(h)).$$

This fixed-vector translation preserves every atom, mixture, and linked type coordinate. The exact unanimity cells are:

Table 9: Unanimity informational rent with hegemonic agenda

Fiber	IR_U^A	Ex ante image
$p = 0$	$(0, \max\{v_U^A(\ell), \beta^2 h\} - v_U^A(h))$	0
$0 < p \le p^*$, $v_U^A(\ell) \ge \beta^2 h$, $\mu^{\text{off}} = 0$	$(0, -(v_U^A(h) - v_U^A(\ell)))$	$-p(v_U^A(h) - v_U^A(\ell))$
Other low-prior fibers	$\emptyset$	$\emptyset$
$p^* < p < 1$, $\mu^{\text{off}} = 0$	$\{(u - v_U^A(\ell), u - v_U^A(h)) : u \in [\max\{v_U^A(\ell), \beta^2 h\}, v_U^A(h)]\}$	$[\max\{v_U^A(\ell), \beta^2 h\} - v_{U,E}^A(p), v_U^A(h) - v_{U,E}^A(p)]$
$p^* < p < 1$, $\mu^{\text{off}} \in (p^*, 1]$	$(v_U^A(h) - v_U^A(\ell), 0)$	$(1 - p)(v_U^A(h) - v_U^A(\ell))$
$p^* < p < 1$, $\mu^{\text{off}} \in (0, p^*]$	$\emptyset$	$\emptyset$
$p = 1$	$(v_U^A(h) - v_U^A(\ell), 0)$	0

Here $v_{U,E}^A(p) = (1-p)v_U^A(\ell) + pv_U^A(h)$. Every existing cell satisfies

$$IR_U^A(\ell) \ge 0, \qquad IR_U^A(h) \le 0,$$

with at least one strict inequality. Majority rent is retained as the exact translation of the full majority payoff correspondence; it has no imposed global sign.

E.10 Public power and information decomposition

For every same-fiber member,

$$\Delta IR^A(o) = IR_U^A(o) - IR_M^A(o) = \Delta V^A(o) - \Delta v^A(o),$$

or equivalently

$$\Delta V^A(o) = \Delta v^A(o) + \Delta IR^A(o).$$

If $\Delta v^A(\ell) < 0$ and a private member has $\Delta V^A(\ell) > 0$, then

$$\Delta IR^A(\ell) > -\Delta v^A(\ell) > 0$$

for that member. Type incidence precedes ex ante aggregation:

$$\Delta V_E^A = \Delta v_E^A + \Delta IR_E^A, \quad \Delta v_E^A = (1-p)\Delta v^A(\ell) + p\Delta v^A(h).$$

Under the sufficient condition in E.5,

$$\Delta V^A(o) \leq -\beta(e/m - \beta h), \qquad \Delta IR^A(o) \leq -\Delta v^A(o) - \beta(e/m - \beta h).$$

This bounds the informational difference relative to the amount required to reverse majority's guaranteed private advantage; it does not assign a sign to $\Delta IR^A(o)$.

The numerical member in Figure 5 gives

$$\Delta IR^A(\ell) = .5480, \qquad \Delta v^A(\ell) = -.4095, \qquad \Delta V^A(\ell) = .1385.$$

These linked values illustrate the identity; they do not calibrate the WTO or select globally from either correspondence.

E.11 Factorial design and identity

At date A, the four cells are

	B	A
public type	$\beta v_g^B(o)$	$v_g^A(o)$
private type	$\beta V_g^B(o)$	$V_g^A(o)$.

Define

$$D_g(o) = v_g^A(o) - \beta v_g^B(o),$$
$$T_g(o) = V_g^A(o) - \beta V_g^B(o),$$
$$I_g(o) = IR_g^A(o) - \beta IR_g^B(o).$$

Substituting $V = v + IR$ in both arms yields, by type and ex ante,

$$T_g = D_g + I_g.$$

The factor β appears once because the no-agenda arm begins one date later. The treatment combines an earlier opportunity with a mandatory proposal. Redating Round 1 to date A in the control would define another structural contrast.

E.12 Direct agenda effects and their institutional difference

The rule-specific public effects are

$$D_U(o) = 1 - \beta$$

and

$$D_M(o) = \begin{cases} v_M^{\text{safe}} - (e/m)\beta^2 o, & o \le 1/m, \\ v_M^{\text{safe}} - \beta o, & 1/m < o < o_M^*, \\ 0, & o \ge o_M^*. \end{cases}$$

The threshold satisfies $o_M^* > 1/m$, but it can be weakly above one, in which case the last branch is empty in the admissible domain. At $o = 1/m$, the first branch applies. The frozen continuation selection creates the jump

$$D_M(1/m) - \lim_{o \downarrow 1/m,\ o > 1/m} D_M(o) = \frac{\beta}{m}\left(1 - \frac{e\beta}{m}\right) > 0.$$

With $\Delta D(o) = D_U(o) - D_M(o)$,

$$\Delta D(o) = \begin{cases} -\beta(e/m)(1-\beta o), & o \leq 1/m, \\ \beta(o - e/m), & 1/m < o \leq o_M^*, \\ 1-\beta, & o \geq o_M^*. \end{cases}$$

The last two formulas coincide at o_M^*. This branchwise statement is the frozen result; no stronger global classification from the later external consultation is promoted here.

E.13 Total private agenda effects

Under unanimity, the exact linked correspondence is

$$T_U = \begin{cases} (1-\beta, \max\{v_U^A(\ell) - \beta^2 h, 0\}), & p = 0, \\ \emptyset, & 0 < p \leq p^*, \\ \{(u - \beta^2 h, u - \beta^2 h) : u \in [\max\{v_U^A(\ell), \beta^2 h\}, v_U^A(h)]\}, & p^* < p < 1,\ \mu^{\text{off}} = 0, \\ \emptyset, & p^* < p < 1,\ \mu^{\text{off}} \in (0, p^*], \\ (1-\beta, 1-\beta), & p^* < p < 1,\ \mu^{\text{off}} \in (p^*, 1], \\ (1-\beta, 1-\beta), & p = 1. \end{cases}$$

In the low-prior interior, the no-agenda control is always empty; the agenda treatment may exist or may also be empty. In the high-prior low-positive off-path fiber, the control exists but the agenda treatment is empty. These different missing arms both propagate to an empty T_U.

Where both arms exist, every unanimity member weakly benefits both types and the maximum gain is $1-\beta$. At $p = 0$, the ex ante gain is $1-\beta$ even if the probability-zero high coordinate is zero.

For majority, retain the exact interaction correspondence:

$$T_M = D_M + I_M.$$

For each coordinate and member,

$$\begin{aligned} T_M(o) > 0 &\iff I_M(o) > -D_M(o), \\ T_M(o) = 0 &\iff I_M(o) = -D_M(o), \\ T_M(o) < 0 &\iff I_M(o) < -D_M(o). \end{aligned}$$

No selection-free sign follows when the source correspondence crosses the threshold. Across rules,

$$\Delta T(o) = \Delta D(o) + \Delta I(o)$$

member by member in every complete product. A robust ranking requires the entire admissible correspondence to lie on one side of $-\Delta D(o)$.

E.14 The diagonal contrast

The diagonal comparison is

$$Q_g(o) = v_g^A(o) - \beta V_g^B(o) = D_g(o) - \beta IR_g^B(o).$$

It compares public information with agenda against private information without agenda, changing two factors at once. It is not a causal effect of agenda or information alone.

Under unanimity,

$$Q_U = \begin{cases} (1-\beta, 1-\beta), & p = 0, \\ \emptyset, & 0 < p \le p^*, \\ (v_U^A(\ell) - \beta^2 h, 1-\beta), & p^* < p \le 1. \end{cases}$$

The ex ante value in the high-prior region is

$$Q_U^E = 1 - \beta - (1-p)\beta^2(h-\ell).$$

Under majority,

$$Q_M = D_M - \beta IR_M^B$$

is the exact image of the applicable linked no-agenda records and remains set-valued where those records do. Q_U may exist where T_U is empty because it does not use the private agenda arm.

Appendix F: Factorization, interaction, and scope

F.1 Exact and economic layers

For majority, the exact signature is

$$\mathrm{Sig}_M^{ex}(R_M) = (\rho, \mu^{\mathrm{off}}, \Lambda_{(\Gamma_\ell, \Gamma_h)}),$$

where Λ codes the diagonal relabeling orbit of the two complete realized laws. Its economic summary is

$$\mathrm{Sum}_M^{econ}(R_M) = (\rho, \mu^{\mathrm{off}}, (q_Z)_\# \Gamma_\ell, (q_Z)_\# \Gamma_h).$$

Unanimity uses the analogous pair Sig_U^{ex} and Sum_U^{econ}. The exact signature preserves the diagonal relabeling orbit of the realized laws and the relevant identities inside those realized records. It does not retain the complete off-path belief, strategy, support, message, or continuation functions. Those functions remain only in the underlying complete binder, and every off-path-sensitive operation must consume that binder. The economic layer retains the anonymous realized laws needed for payoffs, agreement, delay, and anonymous outcomes.

The institutional operator is first applied to complete binders. Only after that product is formed do payoff and outcome maps factor measurably through the economic summaries:

$$\mathcal{C}_{\mathrm{econ}}(R_M, R_U) = \overline{\mathcal{C}}_{\mathrm{econ}}(\mathrm{Sum}_M^{econ}(R_M), \mathrm{Sum}_U^{econ}(R_U)).$$

No Reynolds average is an equilibrium representative. Equality of summaries does not identify named proposals, coalition lotteries, or off-path plans.

For coordinate $r \in \{\ell, h, E\}$, the contrast envelope satisfies

$$\inf \mathcal{C}_r^A = \inf U_r - \sup M_r, \qquad \sup \mathcal{C}_r^A = \sup U_r - \inf M_r.$$

The interval between these numbers is an interval hull, not necessarily an attained set. Likewise, the outcome object is the ordered pair of marginal laws

$$\mathcal{O}_A(\mathbf{d}, \rho, \mu^{\text{off}}) = \left\{((\bar{\Gamma}^\ell_M, \bar{\Gamma}^h_M), (\bar{\Gamma}^\ell_U, \bar{\Gamma}^h_U)) : (R_M, R_U) \in \mathcal{J}^{\text{bind}}_A(\mathbf{d}, \rho, \mu^{\text{off}})\right\}.$$

It is not a joint counterfactual distribution. A common cross-world draw or coordinate splicing would add a primitive absent from the model.

F.2 Interaction with the no-agenda benchmark

Let IR^B_g be the native Round-1 rent vector from the no-agenda benchmark. The exact interaction is the Minkowski difference of complete linked records,

$$I_g = IR^A_g - \beta IR^B_g.$$

For majority, the applicable no-agenda records are the three public regions written in words in Table 6; no general sign is inferred.

Under unanimity:

$$I_U = \begin{cases} (0, \max\{v^A_U(\ell), \beta^2 h\} - v^A_U(h)), & p = 0, \\ \emptyset, & 0 < p \leq p^*, \\ \left\{ \begin{array}{c} (u - v^A_U(h), u - v^A_U(h)) : \\ u \in [\max\{v^A_U(\ell), \beta^2 h\}, v^A_U(h)] \end{array} \right\}, & p^* < p < 1,\ \mu^{\text{off}} = 0, \\ (0,0), & {\scriptstyle p^* < p < 1, \atop \mu^{\text{off}} \in (p^*, 1]}, \\ \emptyset, & {\scriptstyle p^* < p < 1, \atop \mu^{\text{off}} \in (0, p^*]}, \\ (0,0), & p = 1. \end{cases}$$

Thus agenda weakly reduces unanimity's informational rent in every existing high-prior fiber; the change is strict for both types exactly when $\mu^{\text{off}} = 0$ and $u < v^A_U(h)$. At $p = 0$, only the probability-zero high coordinate falls.

The institutional interaction is

$$\Delta I = \Delta IR^A - \beta \Delta IR^B.$$

It exists only when both exact sources exist. It remains set-valued whenever either source does and is never assembled from independently chosen marginal coordinates.

F.3 Payoff dates, existence, and verification limits

All agenda-game values arrive at date A. Native no-agenda Round-1 values are multiplied by β exactly once before comparison. Public agenda payoffs, private agenda payoffs, agenda rents, and their institutional contrasts receive no additional factor.

The existence rules are:

- every public agenda game has a PBE for both rules and types;
- IR_g^A exists exactly where the private agenda source exists;
- ΔIR^A exists exactly where the same-fiber product exists;
- I_g exists exactly where both its agenda and no-agenda rent sources exist;
- ΔI, T_g, and ΔT exist only where every source required by their definitions exists.

An absent source propagates $\emptyset$. It is never encoded as zero, NA, infinity, or a fictional payoff. Mechanical scripts check hashes, schemas, finite identities, inequalities, and enumerated cells. They do not prove PBE completeness, abstract measurability, universal factorization, or the absence of unenumerated deviations; those conclusions come from the textual proofs fixed by the source manifests.

F.4 Causal scope and mandatory treatment

T_g is a structural causal contrast inside the specified model. It is not an empirically identified estimand for the WTO or another institution. The treatment inserts an earlier and mandatory hegemonic proposal stage. Its control starts at Round 1, so T_g includes the value of timing as specified by the game. It does not describe an optional proposal right that H can decline while retaining the same date-A payoff.

When a contrast is set-valued, each comparison is member-specific. A selection-free claim requires the entire exact correspondence to lie on the relevant side of the threshold. We do not promote the six later advisory corollaries to frozen theorems. The exact result remains the source correspondence and its declared domain.

Appendix G: Rationalist crosswalk for the WTO application

Table 10: WTO practices, strategic primitives, and scope

Practice	Strategic translation	Primitive	Qualification and relation to the model
Green Room, Quad, Washington and Brussels meetings	Restricted agenda coalition	Recognition and proposal protocol	Familiar de facto agenda power; removed from the baseline and restored in the extension
Drafting, brackets, secretariat texts, hard-to-amend packages	Control of proposal, amendment rule, or textual reversion	Proposal and amendment protocol	Mostly agenda power under a closed rule; a separate drafting mechanism would require alternative amendment procedures
Market size	Better alternatives and possibly greater patience	$o \in \{\ell, h\}, \beta$	Microfoundation for asymmetric disagreement; patience asymmetry remains outside the model

Practice	Strategic translation	Primitive	Qualification and relation to the model
Absolute reciprocity	Metric for exchanged concessions	Contract metric	Not an independent asymmetry without market-size or adjustment-cost differences
Sanctions, alternative forums, preferential clubs, selective blocking	Credible changes in continuation values	Disagreement payoff and commitment	Microfoundation for outside-option power; a non-credible threat would require another commitment device
Side payments, exceptions, transition periods	Targeted compensation at reservation values	Feasible contract space	Represented by the divisible unit pie; does not determine who keeps the residual
Issue linkage and multidimensional packages	Cross-dimensional compensation	Contract space	Supports the transferable-surplus approximation without making every issue literally divisible
Single undertaking	Linked package and closed acceptance choice	Contract space and closure rule	Distinct from destruction of GATT 1947 as a fallback
U.S. and EC withdrawal from GATT 1947	Lower value of rejecting the WTO package	Disagreement payoffs	Historically specific manipulation of the status quo; not a constitutive feature of the single undertaking

Practice	Strategic translation	Primitive	Qualification and relation to the model
Proposals, objections, working groups, and verification	Observable responses may reveal reservation values	Information structure	Steinberg generally has information flow from weak members to agenda setters; the model reverses that direction
Acquiescence and estoppel	Finality or continuing objection as fixed by the procedure	Voting, amendment, and timing	Not an independent mechanism: any loss of a future right to object must be part of the rule
Retribution, trust, and reputation	Present action changes future beliefs or punishment	Repeated interaction	Outside the two-round model; not inserted as an ad hoc one-shot cost
Formal equality and legitimacy	Consent may change ratification, compliance, or participation	Implementation and domestic politics	Complementary mechanism outside the unitary-actor model
Consensus	Every approval is indispensable; majority supplies substitutes	Approval threshold	The institutional primitive isolated in the baseline

This table is a rationalist reconstruction by the author, not a formal model attributed to Steinberg. It groups observed practices by the strategic object they change. It also preserves three distinctions: acquiescence is part of the voting and amendment protocol, reputation requires future interaction, and the single undertaking does not itself destroy the GATT 1947 fallback.

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